

MesoConnect Atlas Tutorial

Supplementary Methods

Corridor-constrained tractography and along-tract microstructure for mesolimbic pathways

September 2026

<https://diffusientensorimaging-repos.github.io/MesoConnect-Tutorial/>

Introduction

The MesoConnect Atlas is a probabilistic atlas of mesolimbic white-matter pathways derived from 7 T diffusion magnetic resonance imaging data of the Human Connectome Project (HCP; Vu et al., 2015). It covers connections among the ventral tegmental area (VTA), hippocampus, nucleus accumbens, ventral pallidum and amygdala. This tutorial describes how to reconstruct those pathways in new participants using the atlas as an anatomical constraint on participant-level tractography, and how to quantify microstructure along the reconstructed bundles.

Scope

The procedure comprises nine steps. The atlas map for a tract is registered to each participant and dilated into a corridor; probabilistic tractography is run through the corridor from the pathway's seed region to its target; the resulting bundle is cleaned of outlying streamlines; and scalar maps, fractional anisotropy (FA) and the indices of neurite orientation dispersion and density imaging (NODDI; Zhang et al., 2012), are sampled at 100 equidistant points along the bundle. Group-level inference may be performed at three resolutions, the whole tract, quartiles of the tract, or individual nodes, and the tutorial describes the model and correction appropriate to each. Node-wise results may be inspected in the Node-wise Tract Explorer, a browser-based viewer distributed with this site that renders *t*-value profiles, clusters and hemispheric comparisons from a results file.

Two alternative uses of the atlas, averaging a scalar map within the warped atlas mask and sampling along an atlas-derived centerline, are described in the appendix. Corridor-constrained tractography is presented as the primary method because the streamlines are estimated from each participant's own data and support along-tract analysis.

Prerequisites

The tutorial assumes preprocessed diffusion data (denoised, corrected for susceptibility and eddy-current distortion, and brain-masked), a T1-weighted anatomical image, and a Unix environment with FSL (Jenkinson et al., 2012), ANTs (Avants et al., 2008), MRtrix3 (Tournier et al., 2019) and Python with DIPY (Garyfallidis et al., 2014) and pyAFQ (Kruper et al., 2021). Preprocessing is documented in the TUBRIC DTI tutorial; QSIPrep produces equivalent inputs. Estimation of fibre orientation distributions, which that pipeline does not include, is described on the software page.

Organization

Table S1 summarizes the sections of the site.

Table S1

Sections of the Tutorial

Section	Contents
Atlas	File types, tract families, sources of the seed and target regions, downloads
Workflow	The nine steps, each with parameters, verification criteria, the full script, and results from an example dataset
Explorer	The results viewer and its input format
Alternative approaches	Whole-tract extraction; atlas-guided synthetic streamlines
Reference	Consolidated parameters, troubleshooting, software, scripts, suggested methods text, abbreviations, citation, references

Example dataset

Numerical results and figures on the workflow pages are drawn from a single dataset of 57 adults scanned at 3 T with a multi-shell protocol ($b = 1000, 2000, 3250$ and 5000 s/mm²) and processed for the posterior and anterior VTA → hippocampus pathways. The atlas itself was constructed at 7 T. The example dataset is included to illustrate expected output at a field strength typical of most studies.

Atlas overview

The MesoConnect Atlas comprises group probabilistic maps of mesolimbic tracts connecting the ventral tegmental area (VTA), hippocampus, nucleus accumbens, ventral pallidum and amygdala. The maps were generated from Human Connectome Project (HCP) 7 T diffusion data (Vu et al., 2015) using probabilistic tractography (Tournier et al., 2010) with tract-specific exclusion masks, streamline cleaning and group-level probabilistic mapping, and are distributed in Montreal Neurological Institute (MNI) space on the FSL MNI152 1 mm grid ($182 \times 218 \times 182$ voxels).

The atlas represents plausible streamline trajectories. It does not establish monosynaptic connectivity, fibre directionality or neurotransmitter identity, and several components of the hippocampal–VTA circuit are polysynaptic.

File types

Each tract is distributed as three files (Table S2).

Table S2

Atlas File Types

Suffix	Contents	Typical use
*_GroupMean_OverlapProp.nii.gz	Probabilistic map. Each voxel holds the proportion of contributing participants whose binarized tract included it (range 0 to 1).	Visualization, threshold selection, probability-weighted extraction
*_GroupMean_thr50.nii.gz	Binary map of voxels present in at least 50% of participants.	Corridor construction, whole-tract extraction
*_GroupOverlapCount.nii.gz	Count map. Each voxel holds the number of participants whose tract included it.	Reporting participant overlap

The 50% threshold is the default for corridor construction. Higher thresholds (75%, 90%) reduce the map toward a single central trajectory and no longer represent between-participant variation. The 25% threshold retains greater extent and is appropriate when a warped 50% core is short or discontinuous.

Modes of use

Three modes of use are supported.

1. *Corridor-constrained tractography*. The warped and dilated atlas defines the region within which participant-level tractography may travel. Streamlines are estimated from the participant's own data. This mode supports along-tract analysis and is the subject of the workflow section.
2. *Whole-tract extraction*. A scalar map is averaged within the warped atlas mask. This mode is suitable for descriptive summaries; spatial variation along the tract is not retained. See the appendix.
3. *Atlas-guided synthetic streamlines*. A centerline scaffold constructed inside the warped atlas is sampled at 100 nodes. This mode is suitable when participant-level tractography fails in a subset of the sample. The scaffold is a sampling coordinate system rather than a reconstruction. See the appendix.

Coordinate spaces and interpolation

Atlas space is FSL MNI152 1 mm space unless stated otherwise. Native space refers to a participant's T1, diffusion or NODDI grid (NODDI: neurite orientation dispersion and density imaging; Zhang et al., 2012). Releases in MNI 2 mm and MNI152NLin2009cAsym space are planned.

Probabilistic maps are warped with linear interpolation and thresholded in native space. Binary maps are warped with nearest-neighbour interpolation. Spline interpolation is not appropriate for label images. In registration software the moving image is the image being transformed (here the atlas) and the fixed image defines the target grid (the participant); flag names differ between packages. Warped masks should be overlaid on the participant's image and inspected before use.

Interpretive limits

A group atlas warped to native space does not substitute for participant-level tractography when the research question depends on individual anatomy. Dilation increases sensitivity at the cost of specificity, and the smallest dilation that supports reliable tracking should be used. Whole-tract averages are summary descriptors. Synthetic streamlines are sampling scaffolds and should be described as such in reports.

Tract families and regions of interest

The atlas contains 10 tract families (Table S3). Each is processed with the same nine-step workflow; the seed, target and atlas file differ per family. The exclusion masks used during atlas construction are already reflected in each atlas corridor. They are listed here so that users can anticipate where a reconstruction may leak when the corridor is dilated too far. Throughout, the ventral tegmental area is abbreviated VTA and a region of interest is abbreviated ROI.

Table S3

Tract Families in the MesoConnect Atlas

Family	Files (left / right)	Seed	Target	Construction notes
VTA → posterior (body) hippocampus	l_vta_l_hipp, r_vta_r_hipp	VTA	hippocampus	Exclusions: fornix, optic tract and nerve, amygdala, ventral pallidum, accumbens, dorsal striatum, thalamus, cortex and cerebellum, brainstem inferior to the VTA, red nucleus, contralateral hemisphere. A ventral secondary bundle occurs in some participants (see Step 6).
VTA → anterior hippocampus	anterior_l_vta_l_hipp, anterior_r_vta_r_hipp	VTA	hippocampus	Shares its course with the posterior tract for approximately the first 60 to 80 of 100 nodes. Same exclusions.
VTA → amygdala	l_vta_l_amygdala, r_vta_r_amygdala	VTA	amygdala (hippocampus	Optic tract exclusion is required. Fornix

Family	Files (left / right)	Seed	Target	Construction notes
			subtracted)	excluded.
Superior VTA → nucleus accumbens	superior_l_vta_l_accumbens , superior_r_vta_r_accumbens	VTA	accumbens (ventral pallidum subtracted)	Superior and inferior divisions are defined by anterior- commissure inclusion and exclusion masks.
Inferior VTA → nucleus accumbens	inferior_l_vta_l_accumbens , inferior_r_vta_r_accumbens	VTA	accumbens	As above.
Ventral pallidum → VTA	l_vp_l_vta, r_vp_r_vta	ventral pallidum	VTA	Ventral pallidum divided by hemisphere masks.
Hippocampus → nucleus accumbens	l_hipp_l_accumbens, r_hipp_r_accumbens	hippocampus	accumbens	Cutoff 0.08 at 7 T during construction. A QuickBundles division into two clusters was frequently required before cleaning.
Hippocampus → ventral pallidum	l_hipp_l_vp, r_hipp_r_vp	hippocampus	ventral pallidum	As above.
Precommissural fornix	TractSeg fornix $\cap$ hippocampus –accumbens overlap			Derived map, not a corridor target. Currently on an HCP 1.05 mm grid; resample before use.
Postcommissural fornix	TractSeg fornix minus hippocampus–accumbens overlap			As above.

Note. HCP = Human Connectome Project. The example dataset covers the first two families; the downloads page lists the files distributed with this site.

Regions of interest

Seeds and targets are derived from published atlases, thresholded and binarized in Montreal Neurological Institute (MNI) 1 mm space (Table S4). Where a source cannot be redistributed, the region should be rebuilt from the source and the source cited.

Table S4

Sources and Preparation of the Regions of Interest

Region	Source	File stem	Preparation
VTA	7 T probabilistic VTA atlas (Trutti et al., 2021), 25% threshold	left_VTA_0.25_bin, right_VTA_0.25_bin	Red nucleus subtracted where overlapping. The VTA is also subtracted from the lateral hypothalamus and mammillary body exclusion masks.
Hippocampus	Harvard–Oxford subcortical atlas (Frazier et al., 2005; Makris et al., 2006), 50% threshold	HPC_L_0.5_bin, HPC_R_0.5_bin	
Amygdala	Harvard–Oxford subcortical atlas, 50% threshold	amygdala_0.5_bin → left_amygdala_bin, right_amygdala_bin	Hippocampus subtracted; divided by hemisphere masks.
Nucleus accumbens (limbic striatum)	Connectivity-based striatal parcellation (Tziortzi et al., 2014), or FSL accumbens	left_accumbens_bin, right_accumbens_bin	Ventral pallidum subtracted.
Ventral pallidum	Subcortical atlas (Pauli et al., 2018)	ventral_pallidum_bin → left_, right_	Divided by hemisphere masks.
SN/VTA (alternative)	Functional–anatomical SN/VTA masks (Murty et al., 2014)		Comparison resource for the dopaminergic midbrain.

Note. SN = substantia nigra.

Exclusion masks used during atlas construction

The corridor workflow does not require these masks. They are listed so that users know what the atlas already excludes and can reintroduce any of them as an additional `-exclude` argument if a reconstruction enters a specific structure.

- Thalamus; cortical gray matter with cerebellum (single mask); brainstem inferior to the VTA.
- Dorsal striatum: caudate and putamen, with accumbens and ventral pallidum subtracted.
- Lateral hypothalamus and mammillary bodies, with the VTA subtracted.
- Red nucleus, with the VTA subtracted.
- Fornix body, with hippocampus and amygdala subtracted (VTA → hippocampus and VTA → amygdala).
- Optic tract and optic nerve, per hemisphere (VTA → hippocampus and VTA → amygdala).
- Contralateral hemisphere, enforcing ipsilateral tracking.
- Anterior commissure inclusion and exclusion masks, defining the superior and inferior VTA → accumbens divisions.

The region-tidying operations (subtractions and hemisphere divisions) will be distributed as a script with the full atlas release.

Downloads

Files distributed with this site

The ventral tegmental area (VTA) → hippocampus files used for the example dataset are listed in Table S5. All files are in Montreal Neurological Institute (MNI) space on the FSL MNI152 1 mm grid and are binary unless stated otherwise.

Table S5

Atlas and Region Files Distributed With This Site

File	Hemisphere	Contents
left_VTA_0.25_bin.nii.gz	Left	VTA seed (Trutti et al., 2021; 25% threshold)
right_VTA_0.25_bin.nii.gz	Right	VTA seed (Trutti et al., 2021; 25% threshold)
HPC_L_0.5_bin.nii.gz	Left	Hippocampus target (Harvard–Oxford; 50% threshold)
HPC_R_0.5_bin.nii.gz	Right	Hippocampus target (Harvard–Oxford; 50% threshold)
l_vta_l_hipp_1mm_MNI_GroupMean_thr50.nii.gz	Left	Posterior VTA → hippocampus atlas, 50% threshold
r_vta_r_hipp_1mm_MNI_GroupMean_thr50.nii.gz	Right	Posterior VTA → hippocampus atlas, 50% threshold
l_vta_l_hipp_1mm_MNI_GroupMean_OverlapProp.nii.gz	Left	Posterior VTA → hippocampus atlas, probabilistic (not binary)
r_vta_r_hipp_1mm_MNI_GroupMean_OverlapProp.nii.gz	Right	Posterior VTA → hippocampus atlas,

File	Hemisphere	Contents
		probabilistic (not binary)
anterior_l_vta_l_hipp_1mm_MNI_GroupMean_thr50.nii.gz	Left	Anterior VTA → hippocampus atlas, 50% threshold
anterior_r_vta_r_hipp_1mm_MNI_GroupMean_thr50.nii.gz	Right	Anterior VTA → hippocampus atlas, 50% threshold

The following commands retrieve all files listed above into the folder layout of the full release, which is the layout that `00_config.sh` expects (ATLAS_DIR is the MesoConnectAtlas folder).

```
mkdir -p MesoConnectAtlas/roi_maps MesoConnectAtlas/tracts_thresholded_binary_50 \
        MesoConnectAtlas/tracts_probabilistic
cd MesoConnectAtlas
base="https://diffusiontensorimaging-repos.github.io/MesoConnect-Tutorial/atlas"

for f in left_VTA_0.25_bin right_VTA_0.25_bin HPC_L_0.5_bin HPC_R_0.5_bin; do
    curl -sSL "$base/$f.nii.gz" -o "roi_maps/$f.nii.gz"
done
for t in l_vta_l_hipp r_vta_r_hipp anterior_l_vta_l_hipp anterior_r_vta_r_hipp; do
    f="${t}_1mm_MNI_GroupMean_thr50.nii.gz"
    curl -sSL "$base/$f" -o "tracts_thresholded_binary_50/$f"
done
for t in l_vta_l_hipp r_vta_r_hipp; do
    f="${t}_1mm_MNI_GroupMean_OverlapProp.nii.gz"
    curl -sSL "$base/$f" -o "tracts_probabilistic/$f"
done
```

Full release

The complete atlas package will follow the layout below. The remaining families (amygdala, accumbens divisions, ventral pallidum, hippocampus → accumbens, hippocampus → ventral pallidum, fornix divisions), the count maps, the 25% and 75% thresholds, the endpoint maps and the exclusion masks are available from the atlas authors in the interim.

```
MesoConnectAtlas_v1.0/
  Atlases/FSL_MNI152_1mm/
    tracts_probabilistic/          *_GroupMean_OverlapProp.nii.gz
    tracts_thresholded_binary_25/  *_thr25.nii.gz
    tracts_thresholded_binary_50/  *_thr50.nii.gz
    tracts_thresholded_binary_75/  *_thr75.nii.gz
    tracts_overlap_count/          *_GroupOverlapCount.nii.gz
    endpoint_maps/ roi_maps/ exclusion_masks/
  Atlases/HCP_T1_1p05mm/fornix_split_current_space/
  Scripts/ Examples/ Metadata/ Documentation/
```

Source atlases

- Trutti et al. (2021): probabilistic VTA in MNI space; threshold at 25% for a seed region.
- Pauli et al. (2018): subcortical atlas including ventral pallidum, VTA and substantia nigra.
- Harvard–Oxford subcortical atlas (Frazier et al., 2005; Makris et al., 2006), distributed with FSL: hippocampus, amygdala, accumbens, caudate, putamen, thalamus.
- Tziortzi et al. (2014): connectivity-based striatal parcellation; limbic striatum as an accumbens definition.
- Murty et al. (2014): functional–anatomical substantia nigra/VTA masks.
- TractSeg (Wasserthal et al., 2018): fornix and major white-matter tracts.
- MNI152 templates (FSL, `$FSLDIR/data/standard`): the 1 mm brain image and brain mask used for registration.

Workflow overview

The workflow comprises nine steps applied to one tract at a time (Figure S1). Step 1 is performed once per participant and reused across tracts. Steps 2 through 9 are repeated for each tract by changing the tract name and the seed, target and atlas files in the configuration.

Figure S1

Sequence of the Corridor Workflow

1. Step 1. Registration of the MNI template to native T1 space (ANTs SyN).
2. Step 2. Warping of the seed, target and atlas into diffusion space.
3. Step 3. Corridor construction (dilate, add seed and target, invert).
4. Step 4. Cutoff selection by pilot sweep and side-by-side comparison.
5. Step 5. Corridor-constrained tractography (tckgen).
6. Step 6. Bundle cleaning (Mahalanobis outlier removal).
7. Step 7. Quality control (tract-density overlays).
8. Step 8. Along-tract profiles (100 nodes; FA and NODDI).
9. Step 9. Group-level inference (whole tract, quartiles, nodes), with node-wise results viewed in the Node-wise Tract Explorer.

Note. MNI = Montreal Neurological Institute; FA = fractional anisotropy; NODDI = neurite orientation dispersion and density imaging. The Node-wise Tract Explorer is the browser-based results viewer described in the Explorer section.

Inputs

Table S6 lists the per-participant inputs. When the T1 image is already aligned to the diffusion image, as in Human Connectome Project data, no affine matrix is needed: `T1_TO_DWI` is set to `header` in the configuration and the warp script resamples by image header. With the default, `matrix`, a participant without a matrix is reported as missing an input.

Table S6

Per-Participant Inputs

Input	Origin	Path used by the scripts
Skull-stripped T1-weighted image	preprocessing	<code>\$PROJECT/anat/<subj>/<subj>_T1w_brain.nii.gz</code>
Normalized white-matter fibre orientation distribution (FOD; .mif)	MRtrix3 multi-shell multi-tissue CSD and <code>mtnormalise</code>	<code>\$PROJECT/dwi/<subj>/wm_fod_norm.mif</code>

Input	Origin	Path used by the scripts
Diffusion-space brain mask	preprocessing	\$PROJECT/dwi/<subj>/ nodif_brain_mask.nii.gz
Preprocessed diffusion data with <i>b</i> -values and gradient directions (needed only to estimate FODs or fit NODDI)	preprocessing	\$PROJECT/dwi/<subj>/ data.nii.gz, bvals, bvecs
Mean <i>b</i> = 0 image (optional; background of the quality- control images)	preprocessing	\$PROJECT/dwi/<subj>/ mean_b0.nii.gz
T1 → diffusion affine matrix (when T1_TO_DWI=matrix)	FLIRT	\$PROJECT/xfm/<subj>/ str2diff.mat
Scalar maps for profiling	DTIFIT; AMICO NODDI	\$PROJECT/dwi/<subj>/fa.nii.gz; \$PROJECT/noddi/<subj>/fit_*.nii.gz

Note. CSD = constrained spherical deconvolution.

Outputs

```

$OUT/<subj>/
  reg/          mni2t1_0GenericAffine.mat, mni2t1_1Warp.nii.gz, mni2t1_1InverseWarp.nii.gz
  rois/         <tract>_{seed,target,atlas}_{t1,diff}.nii.gz
                <tract>_atlas_dilated.nii.gz  <tract>_inclusion_zone.nii.gz
<tract>_exclusion_mask.nii.gz
  tckgen/<tract>/
                <tract>_pilot_<cutoff>.tck          (Step 4)
                <tract>_<cutoff>.tck  <tract>_<cutoff>_cleaned.tck
$OUT/qc/<tract>_cutoff_pilot/  cutoff comparison panels, cutoff_summary.csv          (Step 4b)
$OUT/qc/<tract>_tckgen_summary.csv  streamlines selected and generated          (Step 5)
$OUT/qc/<tract>/                per-participant overlay images, qc_flags.csv          (Step 7)
$OUT/nodewise/                 <tract>_tract_stats.csv          cleaned-bundle Streamline_count,
                                Mean_length_mm          (Step 6)
                                <tract>_nodewise_all_subjects.csv
                                Subject, Tract, Node, FA, NDI, ODI, FWF          (Step 8)
$OUT/analysis/                 <tract>__<metric>__analysis.csv  one row per participant (Step
8b)
                                <tract>__<metric>__tract_models.csv          (Step 9a)
$OUT/permutation/              <tract>__<metric>__<outcome>_{nodewise,clusters,summary}.csv
                                results_long.csv          Explorer input          (Steps 9b, 9c)
$OUT/logs/                     one log per shell step script

```

A manifest should be kept for each analysis recording the atlas version and threshold, dilation, interpolation, transform files, cutoff, scalar maps, covariates and software versions.

Scripts

All scripts read a single configuration file, `00_config.sh`, which specifies project paths, the participant list, the tract definition and the tractography parameters. The configuration is edited once per project; the tract fields (`TRACT`, `SEED_MNI`, `TARGET_MNI`, `ATLAS_MNI`) are changed for each tract. The shell scripts write a log to `$OUT/logs/`. Steps 0b, 1, 2, 3, 5, 6, 7 and 8a skip participants whose output already exists, so an interrupted run can be restarted with the same command; tractograms and FOD images are written under a temporary name and renamed on completion, so a file left by an interrupted run is never taken for a finished one. Setting `FORCE=1` recomputes existing outputs. Steps 4, 8, 8b and 9 recompute on every run. The Python and R scripts read the settings exported by `source 00_config.sh`; the file must be sourced again after every edit, otherwise those scripts continue to use the previous values. Steps 0b, 1 and 5 require hours on a full sample and should be run under `tmux` or a job scheduler. The concurrency settings in the configuration (`NTHREADS`, `MAXJOBS`, `ANTS_THREADS`, `ANTS_JOBS`, `FOD_JOBS`) were chosen for a 48-core node and should be reduced on a workstation or a shared machine.

```
cd scripts
nano 00_config.sh          # project paths, tract definition, parameters
source 00_config.sh        # the Python and R scripts read these settings

bash 00b_fod_estimation.sh  # only if wm_fod_norm.mif does not yet exist
bash 01_register_mni_to_t1.sh
bash 02_warp_rois.sh
bash 03_build_corridor_mask.sh
bash 04_tune_cutoff.sh "sub-01 sub-02 sub-03 sub-04 sub-05" "0.1 0.08 0.06 0.01"
python 04b_compare_cutoffs.py "sub-01 sub-02 sub-03 sub-04 sub-05" "0.1 0.08 0.06 0.01"
bash 05_tractography.sh
python 06_clean_bundles.py
python 07_visual_qc.py
python 08a_noddi_fit.py      # omit to profile FA only; Step 8 drops metrics with no maps
python 08_node_profiles.py
python 08b_build_analysis_csv.py # analysis files for Step 9

python 09a_tract_models.py --metric NDI --outcomes memory_accuracy
mkdir -p "$OUT/permutation"
Rscript 09b_nodewise_permutation.R "$OUT/analysis/${TRACT}__NDI__analysis.csv" \
    memory_accuracy NDI_ "$OUT/permutation" "${TRACT}__NDI__memory_accuracy"
python 09c_stack_for_explorer.py "$OUT/permutation"
```

The configuration file follows.

The script `00_config.sh` is reproduced in the Supplementary Scripts appendix.

Step 1. Registration of the MNI template to native T1 space

The atlas and region files are defined in Montreal Neurological Institute (MNI) space, whereas tractography is performed in each participant's diffusion space. Moving an MNI-space image into a participant therefore requires two transforms: a nonlinear warp from MNI to the participant's T1 image, computed in this step, and a linear transform from T1 to diffusion space, obtained during preprocessing and applied in Step 2.

A nonlinear registration is necessary because affine transforms account for global rotation, scaling and shear but not for individual differences in the size and position of subcortical structures. Positional errors of a few millimetres in the midbrain displace the ventral tegmental area (VTA) region outside the VTA.

Procedure

Registration is performed with `antsRegistrationSyNQuick.sh` (Avants et al., 2008), with the participant's skull-stripped T1 image as the fixed image and the skull-stripped MNI152 1 mm template as the moving image. Image coverage should match: registering a skull-stripped participant image to a whole-head template biases the solution.

```
antsRegistrationSyNQuick.sh -d 3 \
  -f "$PROJECT/anat/$s/${s}_T1w_brain.nii.gz" \
  -m "$FSLDIR/data/standard/MNI152_T1_1mm_brain.nii.gz" \
  -o "$OUT/$s/reg/mni2t1_" -n 4
```

The outputs are the affine component (`mni2t1_0GenericAffine.mat`), the forward warp from MNI to T1 (`mni2t1_1Warp.nii.gz`, used in Step 2), the inverse warp from T1 to MNI (`mni2t1_1InverseWarp.nii.gz`, used to return participant-level results to atlas space) and the template resampled into participant space (`mni2t1_Warped.nii.gz`).

If the T1 image has not been skull-stripped, this should be done first; atlas construction used SynthStrip (Hoopes et al., 2022) and the example dataset used ANTs brain extraction. If the T1 image is already aligned to the diffusion image, this warp is the only transform required (`T1_TO_DWI=header` in the configuration).

The full script follows. Each participant requires approximately 10 to 15 min with four threads, and the script runs four participants concurrently.

The script `01_register_mni_to_t1.sh` is reproduced in the Supplementary Scripts appendix.

Verification

The script concludes with a per-participant table of the three transform files. Registration quality is assessed by overlaying `mni2t1_Warped.nii.gz` on the participant's T1 image; the ventricles,

corpus callosum and brainstem outline should coincide. A registration that is globally acceptable but locally displaced in the midbrain will manifest in Step 2 as a misplaced VTA region, so this inspection supplements rather than replaces the region-level inspection described there. In the example dataset all 57 participants produced complete transforms.

Step 2. Warping the seed, target and atlas into diffusion space

Three Montreal Neurological Institute (MNI) space images are transformed for each tract: the seed region, the target region and the atlas's 50% binary map. Each passes through the nonlinear warp from Step 1 and then the linear T1 → diffusion transform. Nearest-neighbour interpolation is used at both stages because the images are labels, and the result is re-binarized.

Procedure

```
# MNI -> T1 (ANTs, nonlinear)
antsApplyTransforms -d 3 -i "$ATLAS_MNI" -r "$T1" -o "$d/${TRACT}_atlas_t1.nii.gz" \
  -t "$OUT/$s/reg/mni2t1_1Warp.nii.gz" -t "$OUT/$s/reg/mni2t1_0GenericAffine.mat" -n
NearestNeighbor

# T1 -> diffusion (FLIRT, linear); with T1_TO_DWI=header, -usesqform replaces -init <matrix>
flirt -in "$d/${TRACT}_atlas_t1.nii.gz" -ref "$PROJECT/dwi/$s/nodif_brain_mask.nii.gz" \
  -applyxfm -init "$PROJECT/xfm/$s/str2diff.mat" -out "$d/${TRACT}_atlas_diff.nii.gz" -interp
nearestneighbour
fslmaths "$d/${TRACT}_atlas_diff.nii.gz" -thr 0.5 -bin "$d/${TRACT}_atlas_diff.nii.gz"
```

Two successive nearest-neighbour resamplings of a binary mask have negligible effect on its extent. If the probabilistic map is required in native space, it should be warped with linear interpolation and thresholded afterwards. The full script follows; it completes in seconds per participant.

The script 02_warp_rois.sh is reproduced in the Supplementary Scripts appendix.

Verification

The script ends with a per-participant table of the voxel counts of the warped seed, target and atlas and of the seed–target overlap, which must be zero; participants with missing outputs are listed as such. Voxel counts far from the sample mean (more than two standard deviations is a useful criterion) indicate a failed registration and call for inspection of that participant. Table S7 gives the counts obtained in the example dataset. In that dataset image dimensions, binariness and laterality (left-hemisphere regions located in the left hemisphere) were also checked for every participant, and a cross-correlation below .60 between the warped template and the T1 image was flagged; two participants scored .59 and were retained after visual inspection. These additional checks are not part of the distributed script.

Table S7

Voxel Counts of Warped Regions in the Example Dataset (N = 57), VTA → Hippocampus

Region	Minimum	<i>M</i>	Maximum	<i>SD</i>
VTA (L / R)	32 / 34	44 / 44	56 / 56	5.5 / 5.2

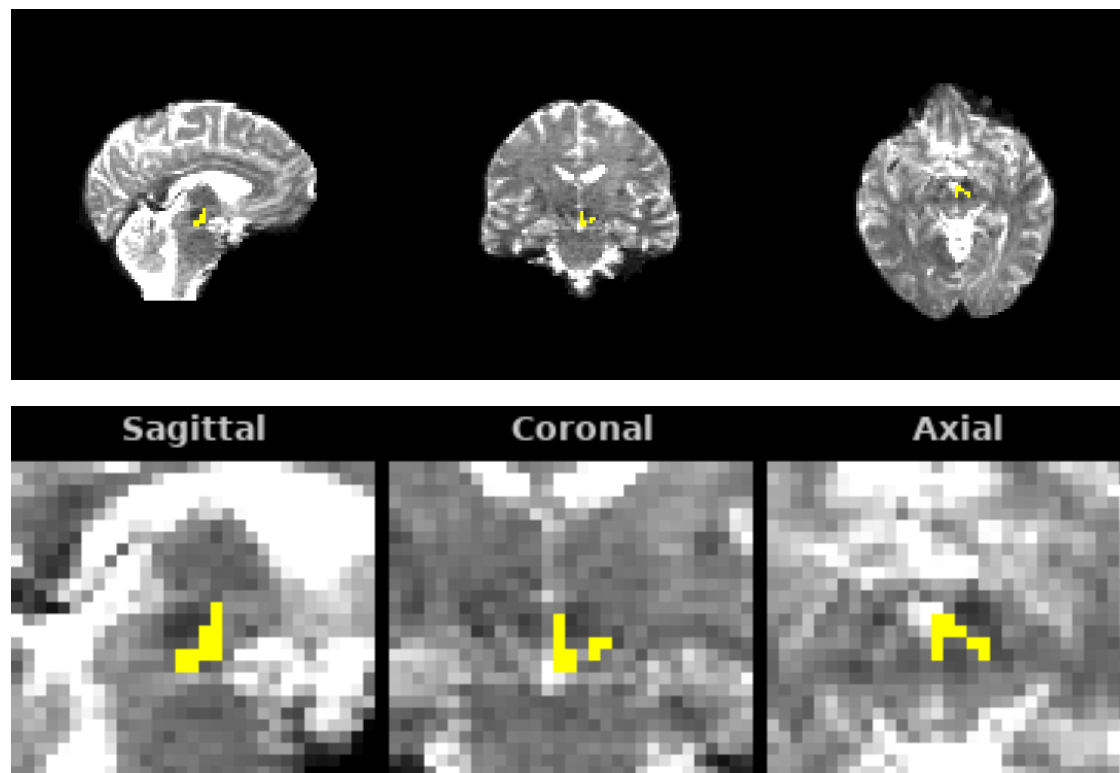
Region	Minimum	<i>M</i>	Maximum	<i>SD</i>
Hippocampus (L / R)	308 / 332	395 / 407	497 / 493	33 / 34
Atlas, 50% (L / R)	107 / 99	126 / 123	146 / 149	9.9 / 10.6

Note. VTA = ventral tegmental area; L = left; R = right.

Visual inspection is performed for every participant by overlaying each warped region on the mean $b = 0$ image. Three views are informative: a whole-brain view for gross placement, a magnified view (5 $\times$) around the region for voxel-level placement, and an orthogonal render for the record (Figures S2 to S4). The VTA should lie in the ventral midbrain anterior to the red nucleus near the midline and occupy a small number of voxels; placement in the cerebral peduncle, pons or outside the brainstem indicates a failed registration. The hippocampus should follow the medial temporal lobe along the floor of the lateral ventricle; overlap with the ventricle or placement in white matter indicates an error. The atlas should form a narrow corridor from the VTA through the midbrain to the target. A region located in ventricle, white matter or cortex, a region with zero voxels, or a left-hemisphere region on the right returns the participant to Step 1.

Figure S2

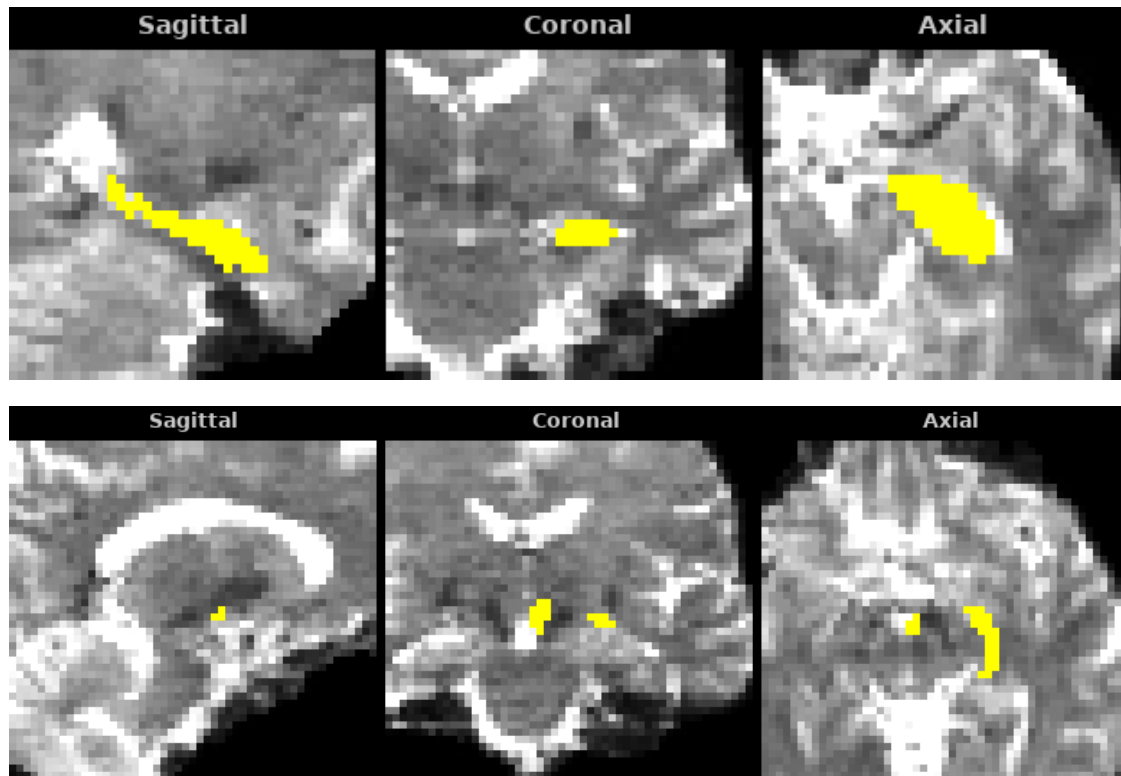
Warped Left VTA Seed Region Over the Mean $b = 0$ Image



Note. Whole-brain view (top) and magnified view (bottom), example dataset.

Figure S3

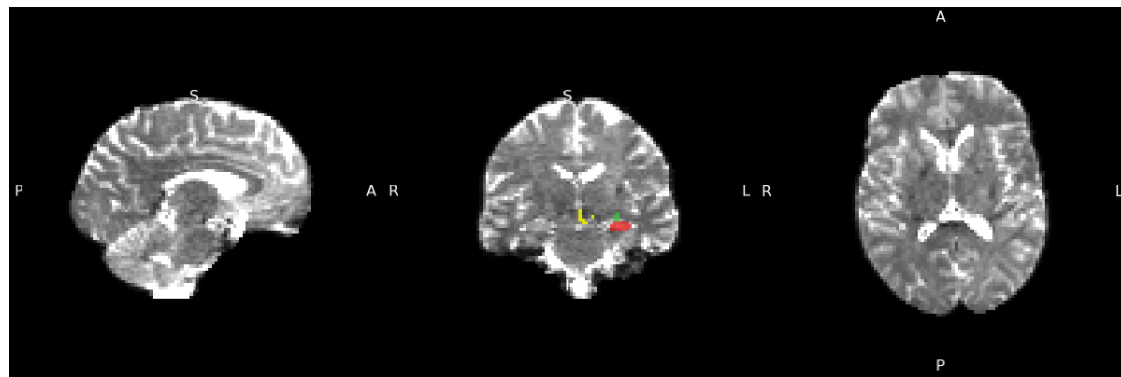
Warped Left Hippocampus Target and Left VTA → Hippocampus Atlas



Note. Hippocampus (top) and atlas at the 50% threshold (bottom), magnified views.

Figure S4

Orthogonal Render of the Warped Regions

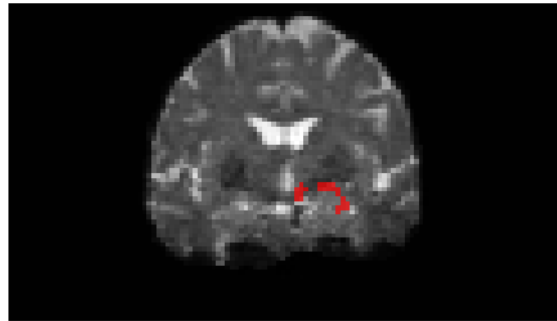
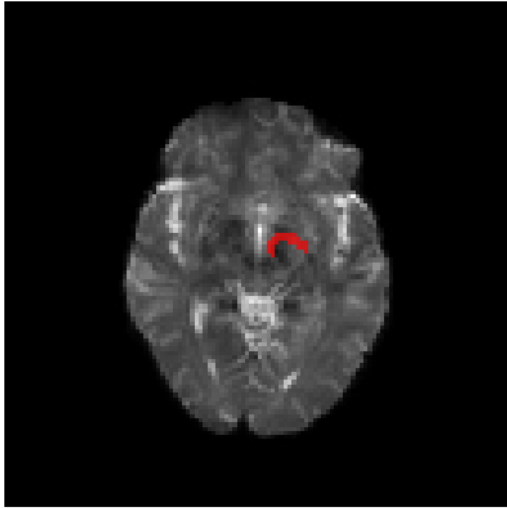


Note. VTA in yellow, hippocampus in red, atlas in green.

For the anterior hippocampus tract the same procedure is repeated with the anterior atlas file; the seed and target regions are shared (Figure S5).

Figure S5

Warped Anterior VTA → Hippocampus Atlas in Diffusion Space



Note. Atlas in red over the mean $b = 0$ image. Left: axial view. Right: coronal view.

Step 3. Corridor construction

Atlas construction used approximately a dozen individually prepared exclusion regions per tract. The corridor workflow replaces them with a single mask derived from the warped atlas. The atlas is dilated, the seed and target regions are added, the union is binarized to form an inclusion zone, and the inclusion zone is inverted. The inverted image serves as the sole exclusion mask for tractography: any streamline leaving the corridor is discarded.

Procedure

```
fslmaths "$d/${TRACT}_atlas_diff.nii.gz" -dilm -dilm "$d/${TRACT}_atlas_dilated.nii.gz"
fslmaths "$d/${TRACT}_atlas_dilated.nii.gz" -add "$d/${TRACT}_seed_diff.nii.gz" \
    -add "$d/${TRACT}_target_diff.nii.gz" -bin "$d/${TRACT}_inclusion_zone.nii.gz"
fslmaths "$d/${TRACT}_inclusion_zone.nii.gz" -binv "$d/${TRACT}_exclusion_mask.nii.gz"
```

The seed and target must be added before inversion; otherwise the corridor terminates short of them and tracking stops at the boundary. The full script follows; the `DILATE_VOX` setting in the configuration controls the number of `-dilm` passes.

The script `03_build_corridor_mask.sh` is reproduced in the Supplementary Scripts appendix.

Dilation

A dilation of two voxels was used in the example dataset at 2 mm isotropic resolution. One voxel is appropriate when registration is accurate and pilot reconstructions appear loose; four voxels serves as a sensitivity setting when registration is uncertain. The 50% atlas is already somewhat wider than the median participant's core, which is why two voxels is generally sufficient. The dilation used should be reported.

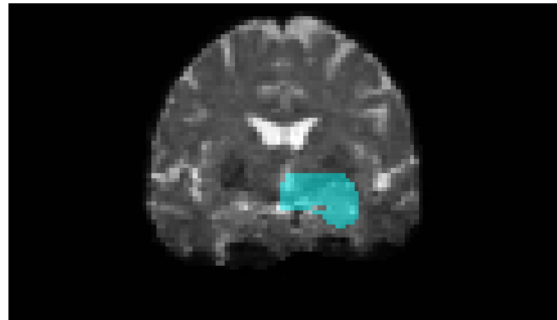
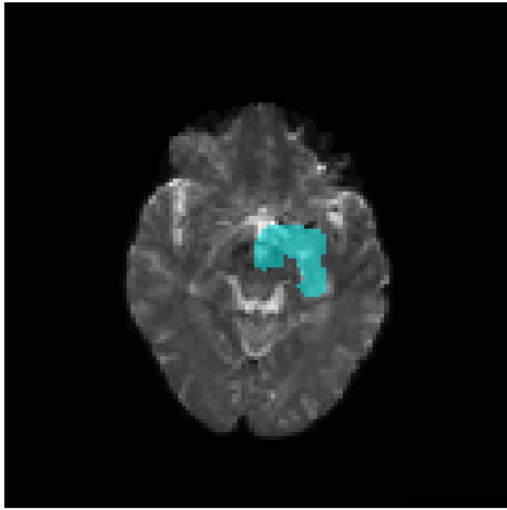
Verification

The script confirms that the seed and target voxels have value 0 in the exclusion mask (that is, are not excluded) and reports the corridor size. Additional checks confirm binariness, agreement of dimensions with the diffusion image, and an inclusion zone below 20% of brain volume. In the example dataset corridor sizes ranged from 1,526 to 1,934 voxels ($M = 1,720$) for the posterior tract and approximately 1,400 voxels for the anterior tract. An automated coverage check that flags inclusion zones below 1% of brain volume flags every participant; the corridor is that small by design.

Visual inspection confirms that the corridor follows a plausible path from seed to target and contains both regions (Figure S6). Contact with the ventricle or extension into cortex indicates excessive dilation or a registration error.

Figure S6

Inclusion Zone for the Anterior VTA $\rightarrow$ Hippocampus Tract



Note. Inclusion zone in cyan over the mean $b = 0$ image, with the seed and target contained within it. Left: axial view. Right: coronal view. VTA = ventral tegmental area.

Step 4. Selection of the FOD amplitude cutoff

The fibre orientation distribution (FOD) amplitude cutoff determines the minimum FOD amplitude at which tracking continues. A high cutoff terminates streamlines before they reach the target; a low cutoff, in unconstrained tracking, produces spurious streamlines. The corridor alters this trade-off. With tracking confined to the corridor, a permissive cutoff produces well-formed bundles while consuming far fewer seeds than a conservative cutoff. This behaviour should be verified for each dataset and tract family by a pilot sweep on a small number of participants, followed by tabulation and side-by-side inspection of the reconstructions.

Cutoff in routine use

The value in routine use with the corridor mask is 0.01, and it is the default in `00_config.sh`. It is the recommended starting value for any tract reconstructed with this workflow; the pilot sweep described below confirms it for a new dataset or tract family and takes under an hour. Table S8 lists the cutoffs that have been used with the atlas.

Table S8

FOD Amplitude Cutoffs in Use

Setting	Cutoff	Outcome
Corridor workflow, 3 T multi-shell data (example dataset): posterior and anterior ventral tegmental area (VTA) → hippocampus	0.01	All 228 runs (57 participants, two tracts, two hemispheres) reached 2,500 streamlines; no participant excluded
Corridor workflow, VTA → accumbens (other users of the atlas)	0.01	Reported to reconstruct the tract
Atlas construction, 7 T: VTA → hippocampus and VTA → accumbens	0.06	Value used to build the atlas
Atlas construction, 7 T: hippocampus → accumbens and hippocampus → ventral pallidum	0.08	Value used to build the atlas
MRtrix3 default for FOD-based tracking	0.05	For reference; not used here

Note. A cutoff of 0.01 is usable only with the corridor's exclusion mask. Without the mask it produces streamlines throughout the brain.

Because lower field strength yields noisier FOD estimates, a cutoff higher than the 7 T values (approximately 0.08) was anticipated for 3 T data. The pilot sweep did not support that expectation: the conservative cutoffs failed to reach the target in several participants, and 0.01

reached it in all of them with about one fifth of the seeds.

Pilot sweep

The sweep uses five participants, four cutoffs and reduced budgets (`-select 1000, -seeds 5000000`); all other parameters match the production run.

```
bash 04_tune_cutoff.sh "sub-01 sub-02 sub-03 sub-04 sub-05" "0.1 0.08 0.06 0.01"
```

The script reports streamlines selected, streamlines generated and mean length for each participant and cutoff. `tcngen` generates one streamline per seed, so the number generated is also the number of seeds consumed. Table S9 gives the streamline counts obtained in the example dataset. At 0.1 most runs exhausted the seed budget before reaching the target. At 0.08 some participants reached the target and one stopped at 309. At 0.06 and at 0.01 all runs reached the target; the 0.01 runs consumed approximately one fifth of the seeds (about 415,000 versus 2.1 million or more at 0.06). Reaching the target in every pilot participant is a necessary condition for a cutoff; the reconstructions must also correspond to the tract, which is assessed next.

Table S9

Streamlines Reached by Cutoff in the Pilot Sweep, Posterior VTA → Hippocampus

Participant	Hemisphere	0.1	0.08	0.06	0.01
P1	L	118	556	1000	1000
P1	R	30	434	1000	1000
P2	L	274	1000	1000	1000
P2	R	165	906	1000	1000
P3	L	1000	1000	1000	1000
P3	R	1000	1000	1000	1000
P4	L	88	309	1000	1000
P4	R	731	1000	1000	1000

Note. Target 1,000 streamlines; seed limit 5 million. P1 to P4 = pilot participants with all four cutoffs completed; the fifth pilot participant was run at 0.06 and 0.01 only and reached the target at both. L = left; R = right.

Side-by-side comparison

For each participant, the tract-density image (TDI) at each cutoff is rendered on the same axial and coronal slices, one column per cutoff. Dice overlap of each cutoff against the most

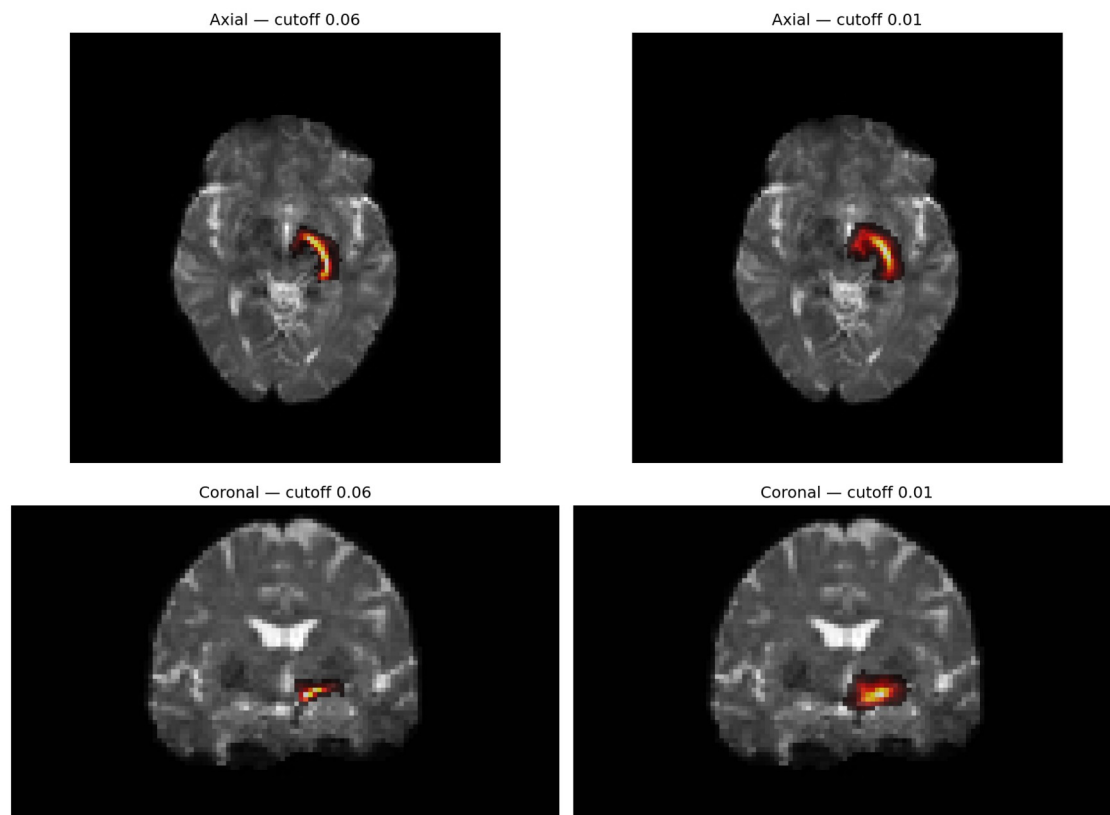
conservative one, the number of TDI voxels, and the mean and standard deviation of streamline length are computed and summarized.

```
python 04b_compare_cutoffs.py "sub-01 sub-02 sub-03 sub-04 sub-05" "0.1 0.08 0.06 0.01"
```

The script writes one panel per participant, `cutoff_summary.csv`, and a summary chart (Figures S7 and S8). In the example dataset the two cutoffs traced the same trajectory in both views. The 0.01 reconstruction was somewhat thicker (mean Dice = .66 against 0.06), mean lengths were similar (approximately 44 versus 47 mm), and the length standard deviation was slightly greater at 0.01 (7 versus 6 mm). Bundle cleaning (Step 6) removes the outlying streamlines responsible for the additional thickness; the cleaned 0.01 bundle is more compact than the uncleaned 0.06 bundle, as shown on that page.

Figure S7

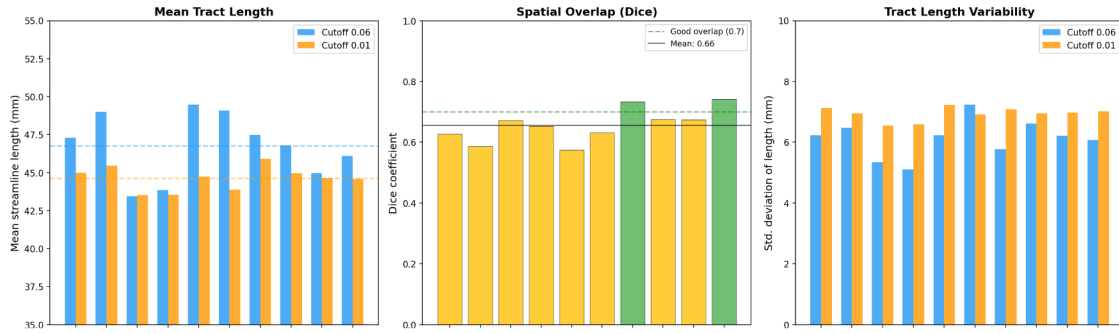
Tract-Density Images at Two Cutoffs, Uncleaned



Note. Cutoff 0.06 (left) and 0.01 (right) for one participant, left posterior VTA → hippocampus.

Figure S8

Summary Statistics by Cutoff Across the Pilot Participants



Note. Bars are grouped by pilot participant and hemisphere. Left: mean streamline length. Middle: Dice overlap between cutoffs. Right: standard deviation of streamline length.

Selection criterion

The cutoff selected is the most permissive value that reaches the streamline target in every pilot participant and whose reconstruction matches the conservative reconstruction. In the example dataset this was 0.01, the value in routine use (Table S8). A higher value is warranted only when the 0.01 reconstruction departs from the conservative reconstruction in the side-by-side images, for example by filling the corridor rather than following the tract; the next value in Table S8 (0.06) is then tried. The sweep should be repeated for each tract family, since a cutoff confirmed for VTA → hippocampus is a starting value for other pathways.

Options not used

Anatomically constrained tractography (ACT) was evaluated during atlas construction and abandoned because the tissue segmentation's white-matter mask was too restrictive relative to the gray-matter mask and tracts did not reconstruct. The `-backtrack` and `-crop_at_gmwmi` options and custom `-angle` and `-step_size` values were left at MRtrix3 defaults. These options should be introduced only in response to a specific, observed failure.

Scripts

The script `04_tune_cutoff.sh` is reproduced in the Supplementary Scripts appendix.

The script `04b_compare_cutoffs.py` is reproduced in the Supplementary Scripts appendix.

Step 5. Corridor-constrained tractography

Tractography is performed with `tckgen` using second-order integration over fibre orientation distributions (iFOD2; Tournier et al., 2010). Seeding is unidirectional from the seed region, the target region is required, and the inverted corridor from Step 3 is supplied as the exclusion mask.

Procedure

```
tckgen "$PROJECT/dwi/$s/wm_fod_norm.mif" "$tck" \  
-algorithm iFOD2 \  
-seed_image "$rois/${TRACT}_seed_diff.nii.gz" -seed_unidirectional \  
-include "$rois/${TRACT}_target_diff.nii.gz" \  
-exclude "$rois/${TRACT}_exclusion_mask.nii.gz" \  
-select 2500 -seeds 25000000 -cutoff 0.01 \  
-minlength 35 -maxlength 65 -stop \  
-nthreads 8 -force
```

Table S10 gives the rationale for each option, and Table S11 the values used per tract family. The script records the number of streamlines selected and the number generated (equal to the number of seeds consumed) per participant in `$OUT/qc/<tract>_tckgen_summary.csv`. The streamline count and mean length that enter the group-level models as covariates are taken from the cleaned bundle in Step 6, because every uncleaned tractogram that reaches the target contains the same number of streamlines. The full script follows. At 3 T with eight threads each participant and hemisphere requires approximately 2 min; 57 participants with both hemispheres required about 4 hr run serially.

Table S10

Tractography Parameters

Option	Value	Rationale
-cutoff	from Step 4 (0.01 in the example dataset)	The corridor permits a low cutoff.
-select	2500	Sufficient for stable profiles and endpoint maps. Streamline count is not a measure of connectivity; participants who stop short are retained.
-seeds	25,000,000	Ceiling on seeding attempts. The default (1,000 × select) is sometimes insufficient to reach 2,500.
-minlength, -maxlength	35, 65 mm	Set for VTA → hippocampus at 7 T. The example dataset retained them at 3 T, with mean

Option	Value	Rationale
		lengths of 44 to 47 mm. Set per tract family from pilot lengths.
-seed_unidirectional	on	Tracking proceeds away from the seed only.
-stop	on	Tracking terminates on entering the target.
gradient table	embedded in the .mif	-fslgrad is omitted when the FOD image was built from a .mif containing gradients.

Note. VTA = ventral tegmental area; FOD = fibre orientation distribution.

Table S11

Parameters by Tract Family

Family	Cutoff with the corridor	Cutoff at 7 T (atlas construction)	Length bounds	Notes
VTA → hippocampus, posterior and anterior	0.01	0.06	35–65 mm	Used for all 57 participants of the example dataset at 3 T.
VTA → amygdala	0.01, confirm by pilot	0.06	determine by pilot	Shorter tract; lower bounds expected.
VTA → accumbens, superior and inferior	0.01	0.06	determine by pilot	0.01 has been used with the corridor by other users of the atlas.
Ventral pallidum → VTA	0.01, confirm by pilot	0.06	determine by pilot	
Hippocampus → accumbens; hippocampus → ventral pallidum	0.01, confirm by pilot	0.08	determine by pilot	Frequently reconstructs as two clusters; see Step 6.

Note. Where a pilot is indicated, the length bounds are taken from the mean lengths reported in the Step 4 summary.

The script 05_tractography.sh is reproduced in the Supplementary Scripts appendix.

Verification

The script reports the number of participants reaching the streamline target. In the example dataset all 114 hemisphere runs reached 2,500 streamlines, consuming 0.5 to 2.0 million seeds (about 4% of the ceiling). The anterior tract also reached 2,500 in all 114 runs, with higher seed consumption (1.5 to 11 million) consistent with its smaller corridor. A participant reaching a lower count, for example 1,800, is retained; a participant reaching a very low count, for example 50, is reviewed in Step 7.

Step 6. Bundle cleaning

Some streamlines within the corridor follow atypical trajectories. Cleaning is performed with `clean_bundle` from pyAFQ (Kruyer et al., 2021), which implements the procedure of Yeatman et al. (2012): each streamline is resampled to 100 points, its Mahalanobis distance from the bundle's mean trajectory is computed, and streamlines exceeding a distance threshold or a length threshold are removed iteratively (Table S12).

Procedure

Table S12

Cleaning Parameters

Parameter	Value	Meaning
<code>n_points</code>	100	Resampling prior to comparison.
<code>clean_rounds</code>	5	Maximum iterations; stops earlier when no streamlines are removed.
<code>distance_threshold</code>	3	Streamlines more than 3 <i>SD</i> (Mahalanobis) from the bundle centroid are removed.
<code>length_threshold</code>	2	Streamlines longer than the mean length by more than 2 <i>SD</i> are removed.
<code>stat</code>	mean	Centroid statistic.
<code>core_only</code>	0	Distance is computed over the full length of each streamline. pyAFQ 3 and later default to the middle 60%; the script sets 0 so that results do not depend on the installed version.

```
from AFQ.recognition.cleaning import clean_bundle
cleaned, keep = clean_bundle(sft, n_points=100, clean_rounds=5,
                             distance_threshold=3, length_threshold=2,
                             stat="mean", core_only=0, return_idx=True)
```

The `core_only` argument exists in pyAFQ 3 and later; the script omits it under earlier versions, whose behaviour it reproduces. In pyAFQ 1.3.2 and earlier the function is imported from `AFQ.segmentation`; the script tries both locations.

The full script follows. It requires pyAFQ and DIPY; on older systems `pip` may require the `zipp` package to be upgraded first.

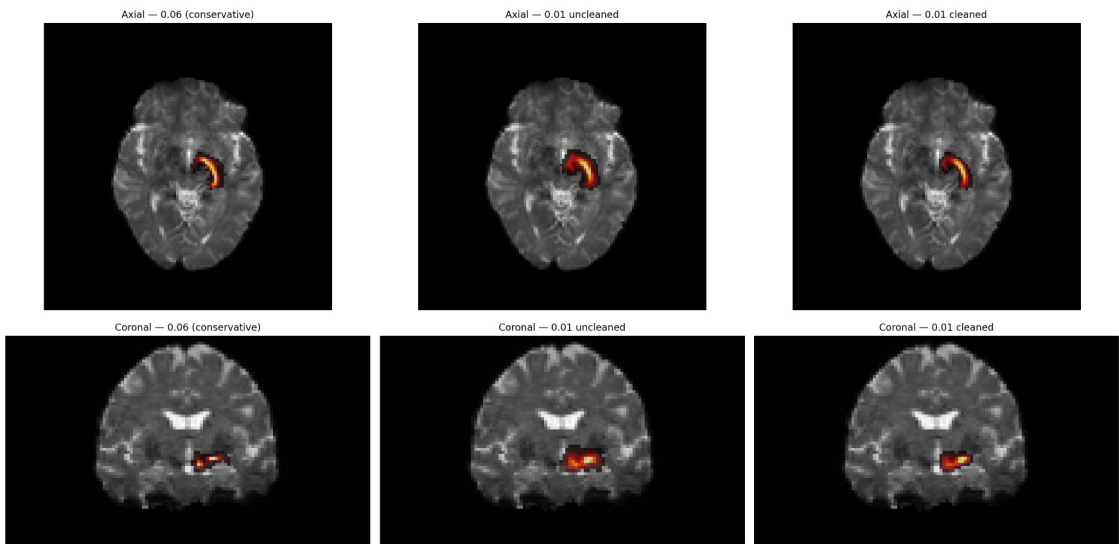
The script `06_clean_bundles.py` is reproduced in the Supplementary Scripts appendix.

Retention

In the example dataset, cleaning retained 26% to 58% of streamlines ($M = 39\%$), with a minimum cleaned count of 659. These values are typical for the thresholds in Table S12, and every participant retained several hundred streamlines. Figures S9 and S10 present the comparison referred to in Step 4: the cleaned 0.01 bundle is more compact than the uncleaned 0.06 bundle, with length standard deviations of approximately 3.5 to 4.5 mm for cleaned bundles against 5 to 7 mm for either uncleaned condition.

Figure S9

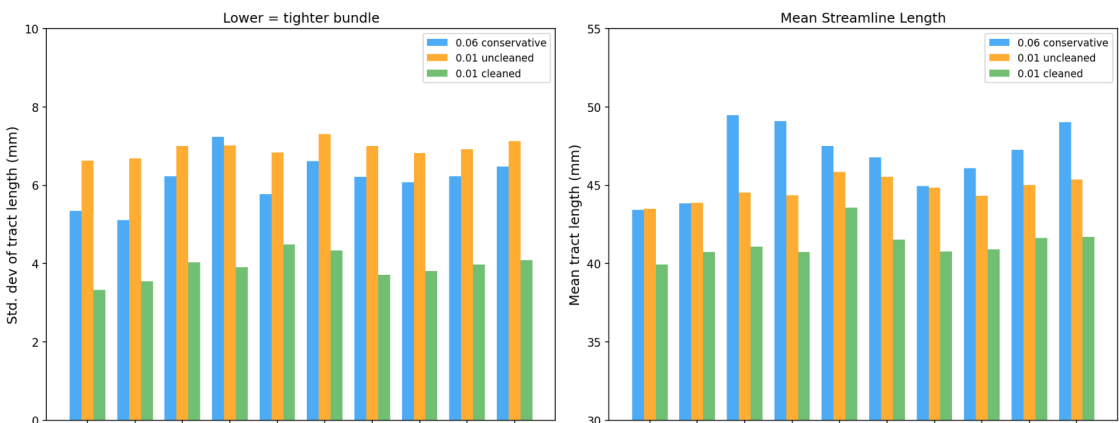
Effect of Cutoff and Cleaning on One Bundle



Note. Left: cutoff 0.06, uncleaned (1,000 streamlines). Middle: cutoff 0.01, uncleaned (2,500). Right: cutoff 0.01, cleaned (824). One participant, left posterior VTA → hippocampus. VTA = ventral tegmental area.

Figure S10

Standard Deviation of Streamline Length by Condition Across the Five Pilot Participants



Note. Bars are grouped by pilot participant and hemisphere. Cleaned 0.01 in green.

Tracts with two bundles

Some tracts reconstruct as two distinct bundles rather than a single bundle with outliers. VTA → hippocampus has a ventral secondary bundle in a subset of participants, and hippocampus → accumbens frequently produces two bundles. Mahalanobis cleaning applied to such a mixture retains the wrong bundle or a blend. The procedure used during atlas construction clusters the streamlines with QuickBundles (Garyfallidis et al., 2012; threshold 5 mm, streamlines resampled to 100 points), cleans the two largest clusters separately, and retains the anatomically correct one after inspection. Setting `QB_SPLIT = True` in the script produces `_cluster1_cleaned.tck` and `_cluster2_cleaned.tck` per participant. The correct cluster is copied to `<tract>_<cutoff>_cleaned.tck` and the script is run again to rebuild the covariate table; the selection should be recorded per participant in the manifest. In the example dataset the corridor and cleaning were sufficient for VTA → hippocampus and the division was not required.

Verification

The cleaned file must exist and contain more than zero streamlines for every participant. Counts before and after cleaning and the retention percentage are reported; retention below 15% or above 80% warrants inspection. The script also writes `$OUT/nodewise/<tract>_tract_stats.csv`, which holds the streamline count and mean length of each cleaned bundle (`Streamline_count`, `Mean_length_mm`). Both index reconstruction quality, vary across participants, and are entered as covariates in Step 9.

Step 7. Quality control of cleaned bundles

Every participant and hemisphere is inspected after cleaning. The script converts each cleaned bundle to a tract-density image (TDI) with `tckmap`, selects the axial and coronal slices of greatest density, overlays the image on the mean $b = 0$ image, and writes one image per participant together with automatic flags (Table S13). Flags appear in the image title and are collected in `qc_flags.csv`.

Table S13

Automatic Flags

Flag	Criterion
LOW_COVERAGE	fewer than 50 TDI voxels
HIGH_COVERAGE	more than 5,000 TDI voxels
LOW_DENSITY	maximum density below 5 streamlines per voxel

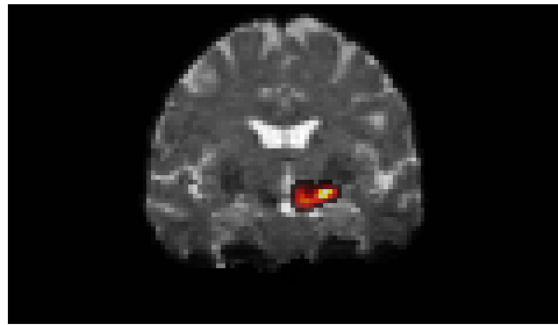
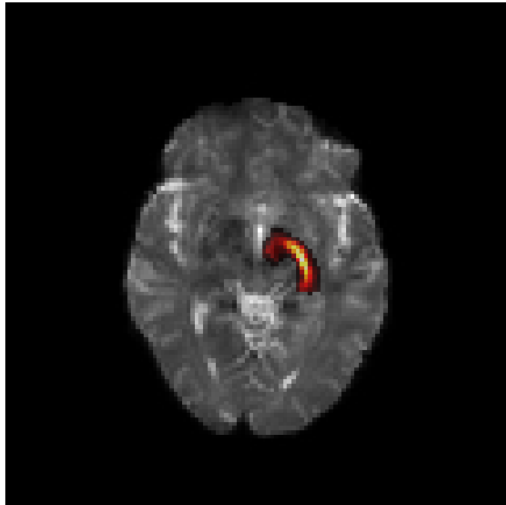
The script `07_visual_qc.py` is reproduced in the Supplementary Scripts appendix.

Criteria

A correct reconstruction shows the expected shape for the tract (for ventral tegmental area [VTA] → hippocampus, an arc from the ventral midbrain laterally into the medial temporal lobe), consistent shape across participants with variation in size, and no isolated voxel clusters, no density in the contralateral hemisphere or frontal lobe, and no empty tracts (Figures S11 to S13).

Figure S11

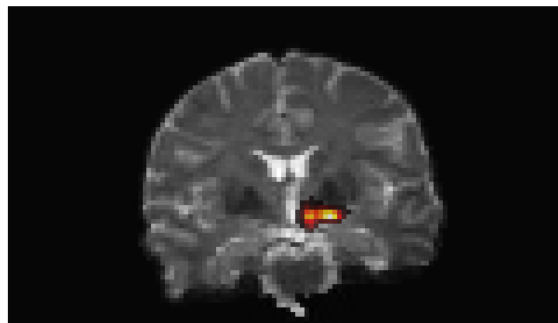
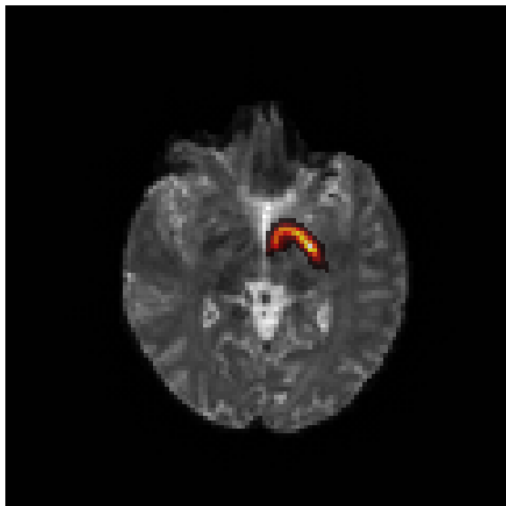
Cleaned Left Posterior VTA → Hippocampus Tract, One Participant



Note. Streamline density over the mean $b = 0$ image. Left: axial view. Right: coronal view. The same layout applies to Figures S12 and S13.

Figure S12

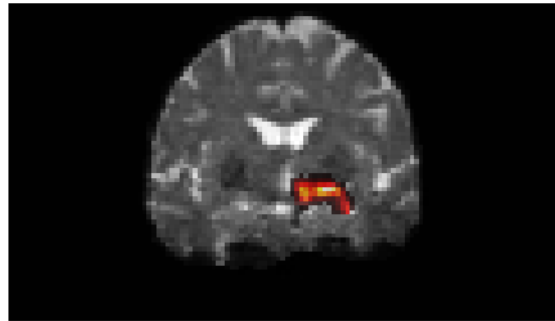
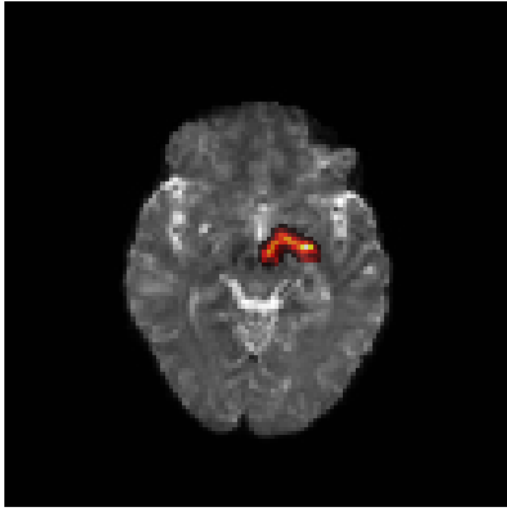
Participant With the Lowest Cleaning Retention



Note. Retention 26%; 659 streamlines.

Figure S13

Cleaned Anterior VTA $\rightarrow$ Hippocampus Tract



Note. Same participant as Figure S11.

Interactive review

The overlay images support a batch review. Flagged participants, and a sample of unflagged participants, should additionally be examined in a viewer with slice navigation, magnification and opacity control.

```
fsleyes "$PROJECT/dwi/$s/mean_b0.nii.gz" \
"$OUT/$s/tckgen/$TRACT/${TRACT}_${CUTOFF}_cleaned.tck" -cm hot -a 70 &

mrview "$PROJECT/dwi/$s/mean_b0.nii.gz" \
-tractography.load "$OUT/$s/tckgen/$TRACT/${TRACT}_${CUTOFF}_cleaned.tck" &
```

Comparison with the atlas

For a quantitative check, each participant's cleaned TDI can be warped to Montreal Neurological Institute (MNI) space with the inverse warp from Step 1 and compared with the atlas by Dice overlap against the 50% map, or by the fraction of the participant tract lying within the 25% probability map. This overlap should not be reported as validation of the atlas, because the reconstruction was constrained by it.

```
tckmap "$tck" "$tdi" -template "$PROJECT/dwi/$s/nodif_brain_mask.nii.gz" -force
fslmaths "$tdi" -thr 1 -bin "${tdi%.nii.gz}_bin.nii.gz"
```

Example dataset

All 114 posterior and 114 anterior tracts passed without flags. The two participants with borderline registration scores in Step 2 passed.

Step 8. Along-tract profiles

Each cleaned bundle is converted to a 100-point profile of a scalar map: fractional anisotropy (FA) from the tensor fit; the neurite density index (NDI), orientation dispersion index (ODI) and free-water fraction (FWF) from neurite orientation dispersion and density imaging (NODDI; Zhang et al., 2012); or any other map on the diffusion grid. Profiling follows the tract-profile approach of Yeatman et al. (2012) as implemented in DIPY (Garyfallidis et al., 2014).

Two operations make profiles comparable across participants. First, because streamlines in a tractogram file may run in either direction, each bundle is oriented against its QuickBundles centroid (Garyfallidis et al., 2012; `threshold=np.inf`, yielding one cluster), and the centroid is itself oriented so that it begins at the end nearer the warped seed region. Node 0 therefore corresponds to the seed end and node 99 to the target end in every participant. Second, `afq_profile` with `gaussian_weights` weights each streamline's contribution at a node by its Mahalanobis distance from the bundle core, reducing the influence of outlying streamlines.

Procedure

```
metric = AveragePointwiseEuclideanMetric(ResampleFeature(nb_points=100))
centroid = QuickBundles(threshold=np.inf, metric=metric).cluster(streamlines).centroids[0]
if distance(centroid[0], seed_centre) > distance(centroid[-1], seed_centre):
    centroid = centroid[::-1] # node 0 at the seed end
oriented = orient_by_streamline(streamlines, centroid)
weights = gaussian_weights(oriented, n_points=100)
profile = afq_profile(data, oriented, affine, n_points=100, weights=weights)
```

The `METRICS` dictionary in the script specifies the scalar maps; the script writes one long-format CSV per tract with one column per metric. Processing 57 participants and four tracts required 20 to 40 min.

The script `08_node_profiles.py` is reproduced in the Supplementary Scripts appendix.

Node range

Nodes near the ends of the profile (0 to 4 and 95 to 99) lie in or adjacent to gray matter and are affected by partial-volume contamination from the seed and target regions. Deep white matter spans approximately nodes 25 to 75. Whether end nodes are trimmed should be decided before modelling.

NODDI inputs

NODDI is fitted with AMICO (Daducci et al., 2015) on the eddy-corrected data; the script `08a_noddi_fit.py`, shown below, performs the fit. The modulated NDI and ODI maps (`fit_NDI_modulated.nii.gz`, `fit_ODI_modulated.nii.gz`), which incorporate the tissue-

weighted partial-volume correction of Parker et al. (2021), are used for profiling; FWF has no modulated form and `fit_FWF.nii.gz` is used. Settings used in the example dataset were `bStep=200` during scheme conversion (rounding jittered *b*-values), `b0_thr=100`, `doSaveModulatedMaps=True`, `doComputeRMSE=True` and `BLAS_nthreads=1`. By default AMICO writes its response kernels to a directory shared by all participants, and concurrent jobs then delete one another's files; the script instead generates the kernels in a working directory for each participant, so several instances can be run at once on different participants (`python 08a_noddi_fit.py sub-01 sub-02`). AMICO also empties its output directory when saving, so nothing else should be stored there.

When NODDI is sampled within a gray-matter region (for example the hippocampus), the model should be refitted with the gray-matter intrinsic parallel diffusivity (approximately 1.1×10^{-3} mm²/s rather than the white-matter default of 1.7×10^{-3} mm²/s) by setting `NODDI_DPAR=1.1e-3`; the script then writes to `$PROJECT/noddi_gm` so that the white-matter maps are retained. The white-matter fit is not appropriate in gray matter and the two fits can yield different results.

The script `08a_noddi_fit.py` is reproduced in the Supplementary Scripts appendix.

Analysis file

Inference in Step 9 reads a wide file with one row per participant: the covariates and outcomes, the streamline count and mean length of the cleaned bundle from Step 6, and the 100 node values as columns `<METRIC>_0` to `<METRIC>_99`. The script `08b_build_analysis_csv.py`, shown below, builds one such file per tract and metric from the long profile CSV, the tract statistics file and a participant-level covariates file (`COVARIATES_CSV` in the configuration; columns `Subject` plus covariates and outcomes).

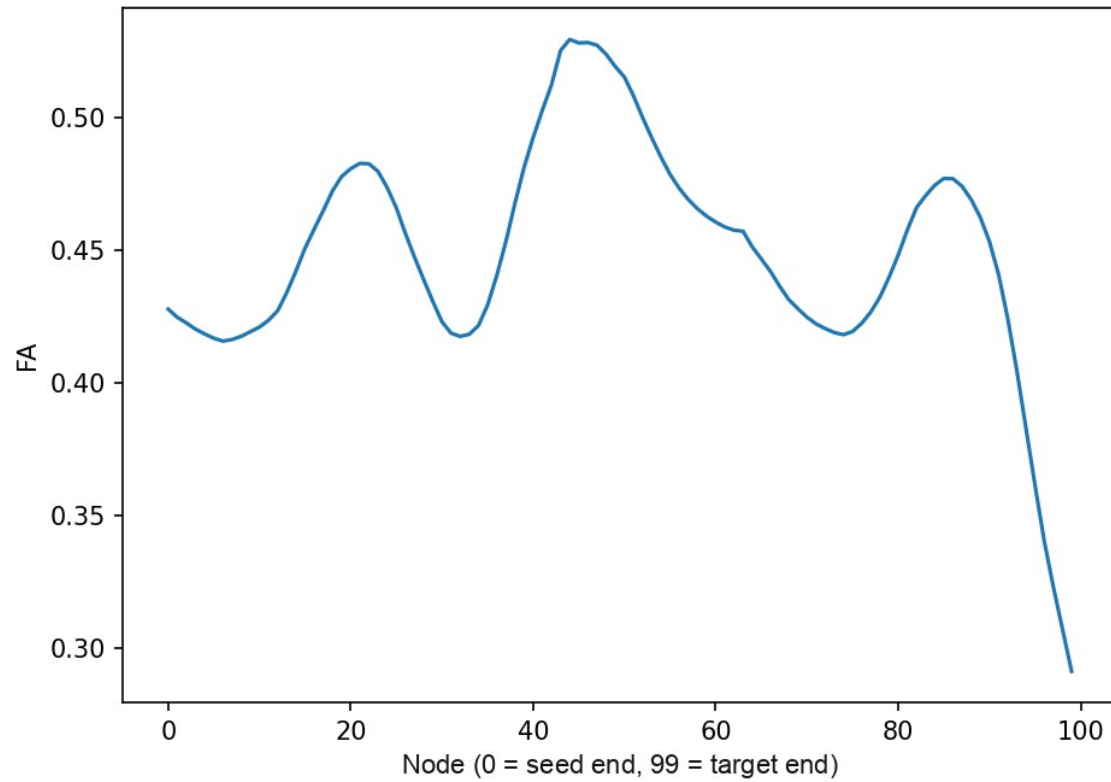
The script `08b_build_analysis_csv.py` is reproduced in the Supplementary Scripts appendix.

Example profiles

Figures S14 to S16 show profiles from one participant in the example dataset.

Figure S14

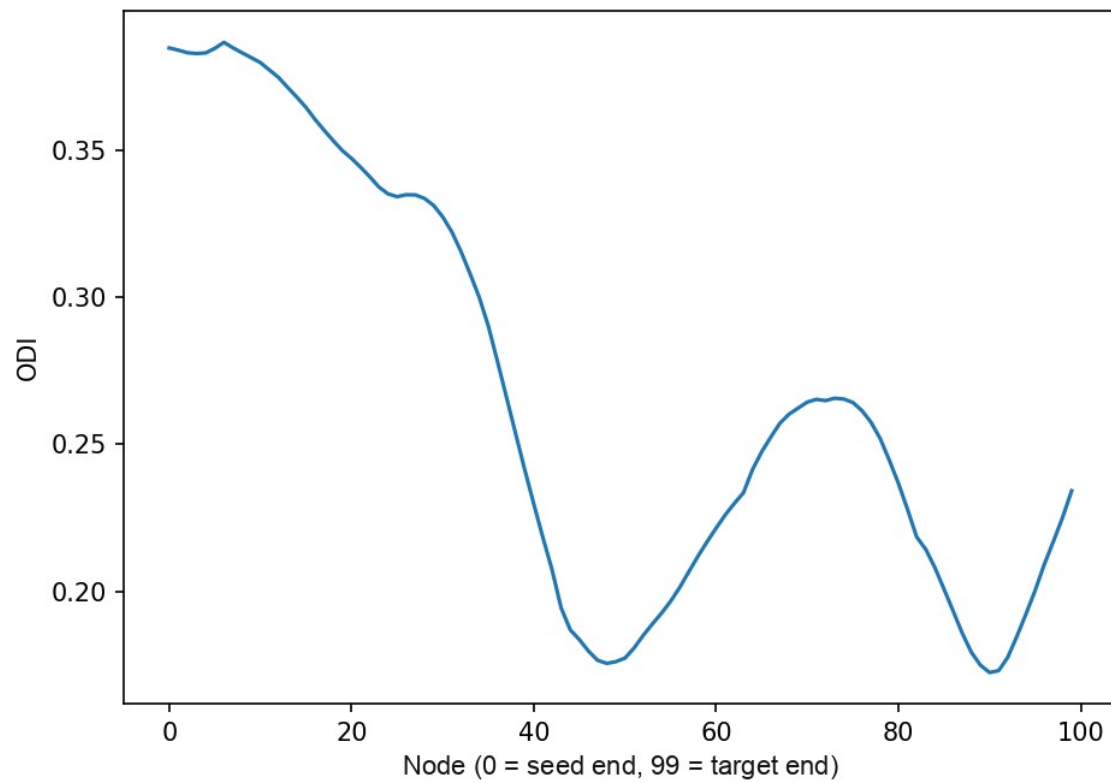
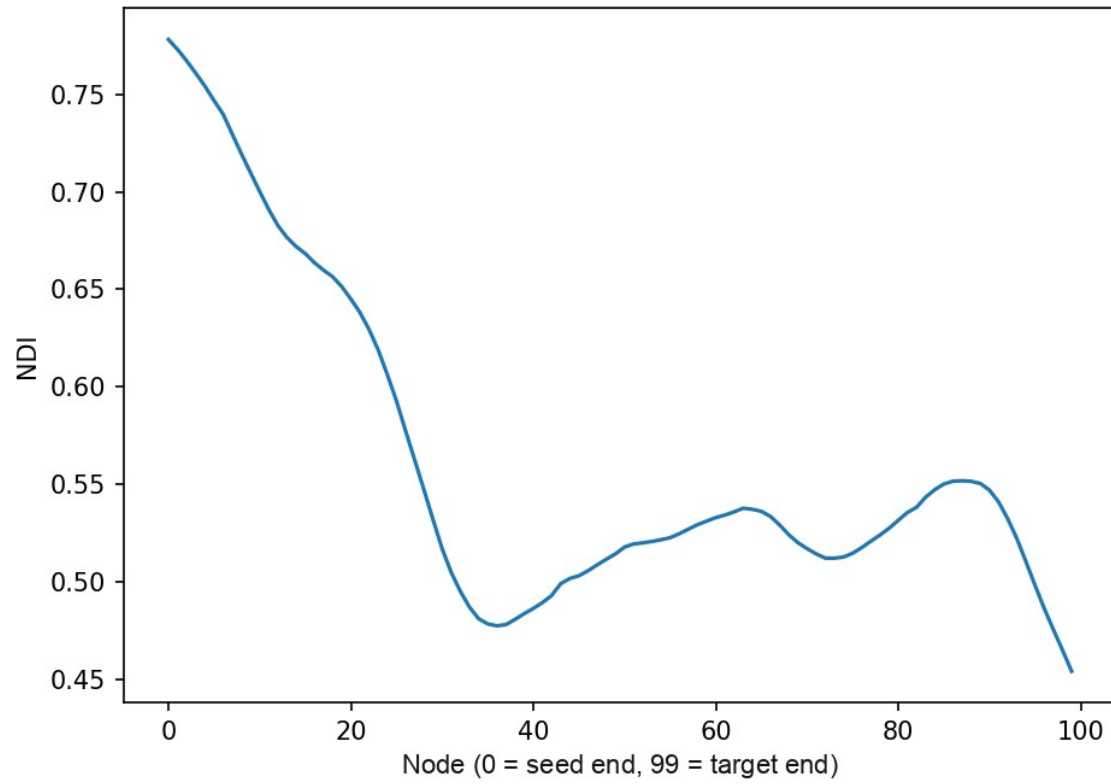
Fractional Anisotropy Along the Left Posterior VTA → Hippocampus Tract

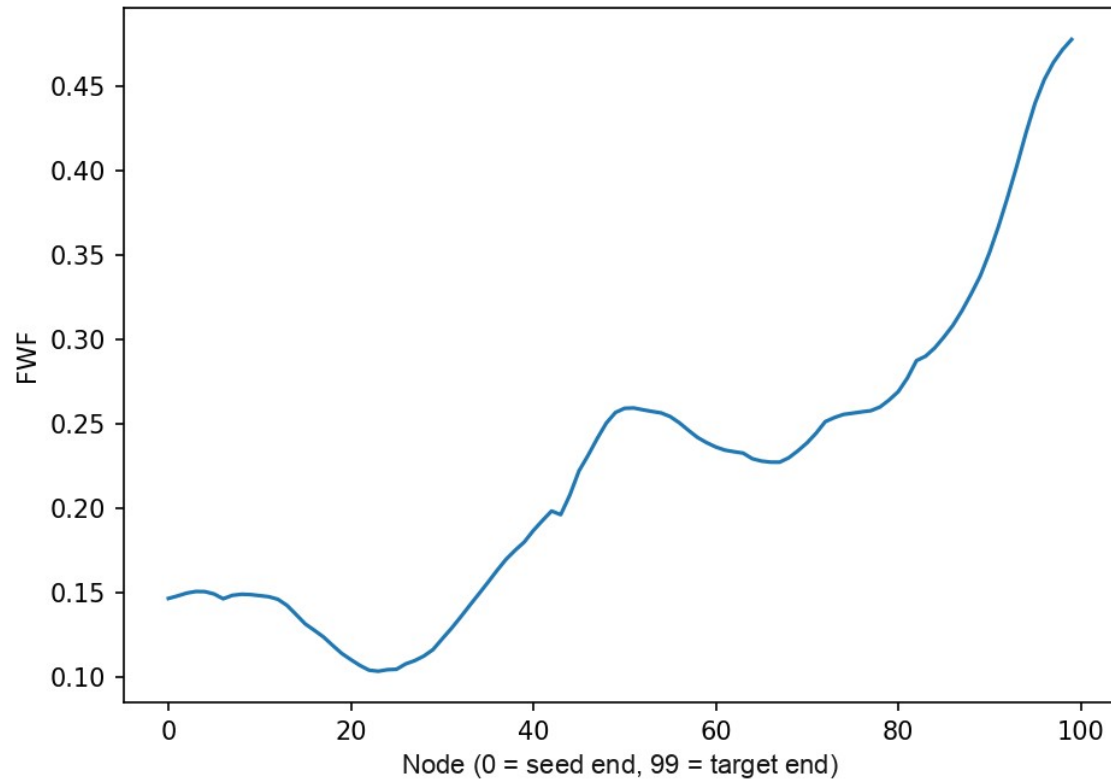


Note. FA peaks in deep white matter and declines toward the hippocampal end. VTA = ventral tegmental area.

Figure S15

NODDI Indices Along the Same Bundle

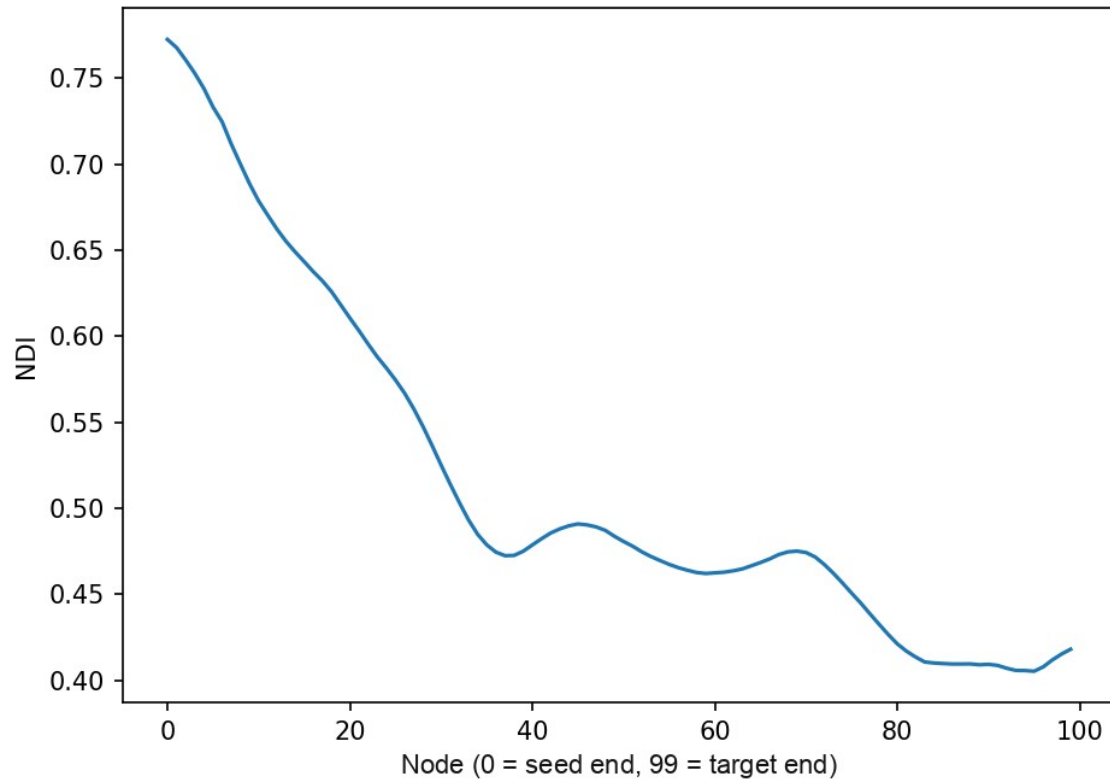




Note. NDI (top), ODI (middle) and FWF (bottom). NDI is high near the VTA and increases again toward the hippocampus; ODI is higher at both ends where fibres fan; FWF rises sharply at the hippocampal end adjacent to cerebrospinal fluid.

Figure S16

NDI Along the Anterior Tract for the Same Participant



Note. The anterior tract shares the early trajectory and diverges late.

Verification

The row count of each CSV should equal the number of participants $\times$ 100. In the example dataset every tract and metric produced 5,700 rows with no skipped participants.

When both hemispheres are processed, alignment should be confirmed before averaging. The across-participant correlation between node i on the left and node i on the right is strongly positive when the profiles are aligned (approximately $r = .98$ in the example dataset) and strongly negative when one side is reversed. Mid-tract averages (nodes 25 to 74) correlated across hemispheres at approximately $r = .80$ to $.90$ for NDI and ODI and $r = .50$ for FA.

Step 9. Group-level inference

Along-tract data can be analysed at three spatial resolutions: the whole tract, quartiles of the tract, and individual nodes (Table S14). The three address different questions, and no single resolution is appropriate for every study. The resolution, or combination of resolutions, should be specified before results are examined, and the same covariates should be used throughout. Where the neurite density index (NDI) is mentioned below, it refers to the neurite orientation dispersion and density imaging metric profiled in Step 8; VTA denotes the ventral tegmental area.

Table S14

Three Resolutions for Along-Tract Inference

Resolution	Unit of analysis	Question	Correction	Script
Whole tract	Metric averaged across nodes; one value per participant per tract	Is the metric associated with the outcome; does the association differ between subregional tracts	None within tract; across outcomes as pre-specified	09a_tract_models.py
Quartiles	Metric averaged within nodes 0–24, 25–49, 50–74, 75–99	Does the association vary along the tract; is it confined to a segment	False discovery rate (FDR) or maximum-statistic permutation across the four quartiles	09a_tract_models.py
Nodes	Metric at each of 100 nodes	Where along the tract is the association located	Cluster-extent family-wise error (FWE) by permutation	09b_nodewise_permutation.R

Whole-tract model

The metric is averaged across all 100 nodes, or across the retained range if end nodes are trimmed (`--trim`), and the outcome is regressed on the averaged metric and the covariates. All

variables are standardized, so coefficients are standardized betas. When two subregional tracts share most of their course (posterior and anterior VTA → hippocampus), both are entered in one mixed model with a subregion term, two rows per participant, a random intercept for participant, and each row carrying its own streamline length and count. The outcome × subregion interaction is tested by likelihood ratio; when it does not improve fit, the main effect is reported. In the mixed model the metric is the dependent variable, because each participant contributes one metric value per tract and a single outcome value. This resolution has the greatest power when the effect is distributed along the tract, and it does not localize the effect.

Quartile model

The same model is fitted with the metric averaged within each quartile. Because adjacent quartiles are correlated, correction across the four should either use the FDR or a maximum-statistic permutation that preserves their dependence. Uniformity along the tract is tested directly by contrasting the seed-end quartile against the target-end quartile with the mean and the difference entered together: when the mean carries the effect and the difference does not differ from zero, the effect is diffuse rather than localized.

Both resolutions are fitted by one script, which reads the analysis files from Step 8b and the covariate list (COVARIATES) from the configuration. With one tract it reports the whole-tract and quartile regressions, the FDR-corrected quartile *p* values and the uniformity contrast; with two tracts it adds the mixed model and the likelihood-ratio test of the interaction for every segment.

```
source 00_config.sh
python 09a_tract_models.py --metric NDI --outcomes memory_accuracy,memory_bias
python 09a_tract_models.py --metric NDI --outcomes memory_accuracy \
    --tracts 1_vta_1_hipp,anterior_1_vta_1_hipp
```

Results are printed and written to \$OUT/analysis/<tract>__<METRIC>__tract_models.csv, one row per model term, with the columns model (ols, uniformity or mixed), tract, segment, outcome, term, n, beta, statistic, p, p_fdr and interaction_p.

Node-wise model

At each node the outcome is regressed on the node's metric and the covariates, and the fit is compared with a reduced model omitting the metric.

```
full      ~ metric_node + ICV + tract_length + streamline_count + motion + age
reduced ~      ICV + tract_length + streamline_count + motion + age
```

Each node yields a *t* statistic and a parametric *p* value, and adjacent nodes with *p* < .05 form clusters. Family-wise error is controlled at the cluster level with a Freedman–Lane permutation procedure (Freedman & Lane, 1983; Winkler et al., 2014): the residuals of the reduced model are permuted, all 100 nodes are refitted, the largest cluster of each permutation is recorded, and the procedure is repeated 5,000 times. A cluster's *p* value is the proportion of permutations whose

largest cluster is at least as long, and a cluster is retained when that value does not exceed .05; the smallest extent that meets this criterion is reported as the extent threshold. The permutations are drawn once from a fixed seed, so results do not depend on the number of cores. Variables are not standardized in this script; the node-wise estimates are in the units of the outcome per unit of the metric.

The script reads the wide analysis file produced in Step 8b and is called once per outcome, tract and metric: `Rscript 09b_nodewise_permutation.R <analysis.csv> <outcome>`
`<METRIC>_ <out_dir> <label>`, with the label written as `<tract>__<metric>__<outcome>`. Covariate column names and the number of permutations are taken from the `COVARIATES` and `N_PERMUTATIONS` settings in the configuration. Outputs per analysis are `_nodewise.csv` (node, estimate, t , p), `_clusters.csv` and `_summary.csv`; each call requires a few minutes on one core.

```
source 00_config.sh
mkdir -p "$OUT/permutation"
for outcome in memory_accuracy memory_bias; do
  Rscript 09b_nodewise_permutation.R "$OUT/analysis/${TRACT}__NDI__analysis.csv" \
    "$outcome" NDI_ "$OUT/permutation" "${TRACT}__NDI__$outcome"
done
```

This resolution localizes an effect along the tract. Its limitation is that cluster-extent correction detects contiguous runs of supra-threshold nodes; when an effect is uniform along the tract, per-node t statistics remain near threshold at every node and the outcome becomes sensitive to small changes in model specification. In that situation node-wise results are descriptive and the whole-tract or quartile model carries the inference.

Covariates

Covariates comprise intracranial volume (ICV), the mean streamline length and streamline count of the cleaned bundle (written by Step 6), absolute head motion and age. Length and count are tract-specific and index reconstruction quality. Streamline count can be strongly correlated with the metric under test ($r = .63$ to $.87$ with tract-mean NDI across the tracts of the example dataset), which reduces the residual variance available to the test. Its inclusion should be decided in advance and both specifications reported.

Choosing a resolution

A hypothesis about a particular segment of a pathway calls for the quartile or node-wise model. A hypothesis about the pathway as a whole calls for the whole-tract model. Exploratory work is often best served by the whole-tract model for inference with the node-wise profile reported as description. Tests across several metrics, tracts and outcomes accumulate rapidly at any resolution; fixing the analytic hierarchy in advance (primary metric, primary outcome family) is more readily justified than correcting across all tests afterwards.

In the example dataset, three social-memory measures were associated with NDI in the whole-

tract model with no subregion interaction. In the quartile model one of them (positivity bias in false memories) was significant in all four quartiles and the seed-end versus target-end contrast was null for all three, indicating uniform effects. Node-wise clusters were located at nodes 27 to 55 and changed with covariate specification. The results were therefore reported with the whole-tract model as primary, the quartile model as confirmation of uniformity, and node-wise profiles as description. Applied to the analysis files of that dataset, `09a_tract_models.py` reproduces the reported whole-tract and quartile estimates.

Preparing node-wise results for the Explorer

Node-wise outputs are stacked into a single long-format file for the Node-wise Tract Explorer, the browser-based viewer described in the Explorer section.

```
python 09c_stack_for_explorer.py "$OUT/permutation" # writes results_long.csv in that directory
```

The file contains one row per node with the columns `outcome`, `tract`, `metric`, `node`, `t`, `p`, together with `hemisphere`, `N`, `covariates`, `extent_threshold`, `cluster_p` and `passed`.

Scripts

The script `09a_tract_models.py` is reproduced in the Supplementary Scripts appendix.

The script `09b_nodewise_permutation.R` is reproduced in the Supplementary Scripts appendix.

The script `09c_stack_for_explorer.py` is reproduced in the Supplementary Scripts appendix.

Node-wise Tract Explorer

The Node-wise Tract Explorer is a browser-based viewer for node-wise statistical results distributed with this site. It reads a results file in the browser, without uploading it, and presents each analysis as a table row that expands to show the node-wise t -value profile, the clusters identified, and, where both hemispheres are present, a left–right comparison. Whole-tract and quartile results are tabular and do not require it.

Open the Explorer · [sample file](#) · [example dataset](#)

Input format

The input is a long-format CSV with one row per node. Six columns are required: `outcome`, `tract`, `metric`, `node`, `t` and `p`. Optional columns are `hemisphere` (L or R), `N`, `covariates`, `extent_threshold`, `cluster_p` and `passed`. Any additional column is treated as a grouping variable and becomes a filter. The column names produced by `09b_nodewise_permutation.R` (`Node`, `t_value`, `p_value`) are accepted without renaming. `09c_stack_for_explorer.py` (Step 9) produces this file from the permutation outputs.

Derived quantities

From the node rows the viewer derives the number of significant nodes, clusters (contiguous runs with $p < .05$), the largest cluster, which clusters exceed the extent threshold, and, when a hemisphere column is present, the counts of significant nodes per hemisphere and their overlap.

Display

Selecting a row shows the t -value profile with a labelled axis and reference lines at ± 2 , significant nodes highlighted, retained clusters shaded, and a cluster table. For lateralized data the left and right profiles are drawn on a common axis so that both hemispheres' full t -value profiles are visible. The hemisphere panel is descriptive: it reports the number of significant nodes per side and their overlap and does not constitute a test of lateralization.

Page labels may be supplied in the URL, for example `?title=...&method=Freedman-Lane&node0=VTA&node1=hippocampus&n_perms=5000` for a ventral tegmental area (VTA) → hippocampus analysis, and `?data=URL` loads a hosted file directly.

Whole-tract extraction

In this approach an atlas map is warped into a participant and a scalar map is averaged within it. It is suitable for descriptive summaries, replication checks and visualization. It is not recommended as a primary endpoint because spatial variation along the tract is not retained.

Procedure

The 50% binary map provides a conservative core; the probabilistic map supports probability-weighted extraction. The map is warped to T1 space with the Step 1 transforms, using nearest-neighbour interpolation for binary maps and linear interpolation followed by thresholding for probabilistic maps.

```
antsApplyTransforms -d 3 -i "$ATLAS_PROB" -r "$T1" -o prob_in_T1.nii.gz \
  -t mni2t1_1Warp.nii.gz -t mni2t1_0GenericAffine.mat -n Linear
fslmaths prob_in_T1.nii.gz -thr 0.50 -bin prob50_in_T1_bin.nii.gz
```

The mask is then moved to the scalar map's grid. When the T1 and the scalar map are header-aligned and differ only in grid, the mask is resampled (`flirt -in mask -ref scalar -applyxfm -usesqform -interp nearestneighbour`); otherwise the T1 → diffusion transform is applied. Voxel count, volume and the mean and standard deviation of the scalar within the mask are then extracted.

```
fslstats mask_on_scalar_grid.nii.gz -V # voxel count, mm3
fslstats "$NDI" -k mask_on_scalar_grid.nii.gz -M -S # mean, SD
```

Reporting

For multi-shell data with a fitted neurite orientation dispersion and density imaging (NODDI) model, the neurite density index (NDI) is generally preferred to fractional anisotropy (FA) as a whole-tract summary. It remains a model-derived quantity sensitive to partial volume, registration and mask boundaries, and should be reported with covariates for age, sex, motion, intracranial volume and whole-white-matter NDI and described as a summary measure.

Atlas-guided synthetic streamlines

This approach is intended for studies that require along-tract profiles but cannot reconstruct the tract in every participant. The warped atlas serves as a geometric scaffold: an ordered centerline is estimated from one endpoint to the other within the mask, 100 nodes are placed along it, jittered streamlines may be generated within the mask, and scalar maps are sampled at each node. The resulting streamlines constitute a reproducible sampling coordinate system rather than a tractography reconstruction, and the approach is less established than corridor-constrained tractography.

Procedure

The probabilistic map is warped onto the scalar grid with linear interpolation and thresholded, at 50% for a conservative core or 25% when the core is short or discontinuous; lower thresholds require justification. The scaffold is built from the mask, the probability map, and native-space start and end regions, with node 0 assigned to the seed side and node 99 to the target side for every participant; an image of the node ordering is saved. Each scalar map is sampled at every node using pyAFQ, DIPY or a custom function, producing one row per participant, tract, metric and node in the long format read by the Explorer. Modelling proceeds as in Step 9 of the main workflow, with cluster or permutation correction.

Reporting

Recommended language: "Atlas-guided streamline scaffolds were used to define a consistent along-tract coordinate system for node-wise sampling." The scaffold should not be described as participant-specific tractography.

Parameters

Table S15 consolidates the parameters used in atlas construction at 7 T and in the example dataset at 3 T.

Table S15

Consolidated Parameters

Step	Parameter	Atlas construction (7 T)	Example dataset (3 T)	Notes
Atlas	threshold	50%	50%	25% when a wa core is short or discontinuous
Registration	tool	antsRegistrationSyNQuick.s h	same	Skull-stripped participant imag against MNI152_T1_1mm n
Warping	interpolation	nearest-neighbour	same	Linear for proba maps, then thres
Corridor	dilation	1–2 voxels	2	4 when registrat uncertain
Tractography	algorithm	iFOD2 (default)	same	
Tractography	-cutoff	0.06 (VTA → hippocampus); 0.08 (hippocampus → accumbens)	0.01	0.01 is the value routine use with corridor mask an default in 00_config.sh; confirm by pilot 4). MRtrix3 def 0.05
Tractography	-select	2500	2500	
Tractography	-seeds	25,000,000	25,000,000	
Tractography	-minlength, -maxlength	35, 65 mm	35, 65 mm	Per tract family
Tractography	-seed_unidirectional	on	on	

Step	Parameter	Atlas construction (7 T)	Example dataset (3 T)	Notes
	, -stop			
Tractography	ACT, backtrack, angle, step size	off / default	off / default	ACT did not reconstruct with masks
Cleaning	n_points, clean_rounds	100, 5	same	
Cleaning	distance_threshold, length_threshold	3, 2	same	Mahalanobis SD length SD
Cleaning	QuickBundles threshold	5 mm	not used	For two-bundle
Quality control	flags		voxels < 50 or > 5,000; maximum density < 5	
Profiling	nodes	100	100	End nodes 0–4 and 99 may be trimmed
Profiling	weighting	Gaussian (Mahalanobis)	same	
NODDI	intrinsic parallel diffusivity, white / gray matter	$1.7 / 1.1 \times 10^{-3} \text{ mm}^2/\text{s}$	same	Refit for gray-matter regions
NODDI	maps	modulated NDI and ODI; FWF	same	
Inference, whole tract	model		mixed model with subregion term; likelihood-ratio test for interaction	one value per participant per tract
Inference, quartiles	segments and correction		nodes 0–24, 25–49, 50–74, 75–99; FDR or maximum-statistic permutation	seed-end versus end contrast for uniformity
Inference, nodes	permutations	5,000	5,000	Freedman–Lane
Inference,	cluster criterion	extent $\geq$ 95th percentile of null	same	$\alpha = .05$ per node

Step	Parameter	Atlas construction (7 T)	Example dataset (3 T)	Notes
nodes		maximum		
Inference, nodes	covariates		ICV, length, count, motion, age	Count is collinearity; NDI; decide in advance

Note. VTA = ventral tegmental area; ACT = anatomically constrained tractography; NODDI = neurite orientation dispersion and density imaging; NDI = neurite density index; ODI = orientation dispersion index; FWF = free-water fraction; FDR = false discovery rate; ICV = intracranial volume.

Troubleshooting

Table S16 lists common problems, their likely causes and remedies. A quality-assurance checklist follows.

Table S16

Common Problems, Causes and Remedies

Symptom	Likely cause	Remedy
Warped atlas displaced anteriorly or posteriorly	Poor T1 → MNI registration, or mismatched coverage (skull-stripped participant image with a whole-head template)	Check brain extraction; use the brain template; overlay the warped template on the T1 before warping labels
Left and right appear reversed	Incorrect transform, header orientation error, mislabelled hemisphere	Check with <code>fslorient</code> and <code>fslhd</code> ; overlay known asymmetric anatomy
Atlas mask empty after warping	Threshold too strict after linear interpolation, or grid mismatch	Inspect the probability map before thresholding; use the 25% map; warp the binary map with nearest-neighbour interpolation
Mask includes ventricle or gray matter	Registration error or excessive dilation	Reduce dilation; inspect in native space
VTA in the cerebral peduncle or pons	Local registration failure	Rerun registration for the participant; consider a midbrain-weighted mask via <code>-x</code>
<code>tckgen</code> reaches far fewer streamlines than the target	Cutoff too high for the field strength; corridor too narrow; seed or target disconnected from the corridor	Run the pilot sweep; confirm the seed and target lie within the inclusion zone; try the 25% atlas or one additional voxel of dilation
Tract appears thick at a permissive cutoff	Expected before cleaning	Compare cleaned bundles; if still thick, reduce dilation
Cleaning removes nearly all streamlines	Two bundles cleaned as one, or a sparse raw tract	Apply the QuickBundles division; check the raw streamline count
Profile endpoints	Orientation failed	Assign node 0 by the seed region;

Symptom	Likely cause	Remedy
reversed in some participants		check centroid orientation per participant
Whole-tract NDI implausibly high or low	Grid mismatch, partial volume, global tissue differences	Confirm a common grid; add a whole-white-matter NDI covariate; inspect the overlay
AMICO <code>FileNotFoundError</code> during parallel runs	Kernel directory regenerated by every job	Generate kernels once on one participant before starting the pool
<code>pip install pyAFQ</code> fails on zipp	Outdated system package	<code>pip install --user --upgrade zipp</code>
Coverage audit flags every corridor as too small	Corridors are below 1% of brain volume by design	Raise the threshold or disregard the flag

Note. MNI = Montreal Neurological Institute; VTA = ventral tegmental area; NDI = neurite density index.

Quality-assurance checklist

- Record atlas version, threshold, dilation, interpolation, transforms, cutoff, scalar maps, covariates and software versions.
- Inspect the atlas in MNI space before warping.
- Inspect every warped region over the T1 and diffusion images.
- Check voxel counts and volumes after warping.
- Save at least one overlay image per participant and step.
- Inspect streamlines in native space and compare with the atlas in MNI space.
- Verify node ordering and coverage.
- Check missingness by tract and participant before group models.

Software

Table S17 lists the software used in the workflow. The workflow depends on FSL (Jenkinson et al., 2012), ANTs (Avants et al., 2008), MRtrix3 (Tournier et al., 2019) and Python with DIPY (Garyfallidis et al., 2014) and pyAFQ (Krupe et al., 2021); AMICO (Daducci et al., 2015) is required only for neurite orientation dispersion and density imaging (NODDI) fitting and R only for the node-wise permutation test.

Table S17

Software Used in the Workflow

Software	Use
FSL	Templates; <code>fslmaths</code> , <code>fslstats</code> , <code>flirt</code> ; <code>probtrackx2</code> for the FSL route
ANTs	<code>antsRegistrationSyNQuick.sh</code> , <code>antsApplyTransforms</code>
MRtrix3	<code>mrconvert</code> , <code>dwi2response</code> , <code>responsemean</code> , <code>dwi2fod</code> , <code>mtnormalise</code> , <code>tckgen</code> , <code>tckinfo</code> , <code>tckstats</code> , <code>tckmap</code> , <code>mrview</code>
Python 3 with DIPY, pyAFQ, nibabel, numpy, scipy, pandas, statsmodels, matplotlib	Cleaning, profiling, quality control, whole-tract and quartile models
AMICO	NODDI fitting
R with readr, dplyr, stringr, tibble, foreach, doParallel	<code>09b_nodewise_permutation.R</code>
FSLeyes, mrview or ITK-SNAP	Inspection
TractSeg (Wasserthal et al., 2018; optional)	Fornix and control tracts

Fibre orientation distribution estimation

When preprocessing ends at the tensor fit, as in the TUBRIC pipeline, the following MRtrix3 commands produce the normalized white-matter fibre orientation distribution (FOD) image required by the workflow. Response functions are estimated per participant with the unsupervised method of Dhollander et al. (2016), averaged across participants, and used for multi-shell multi-tissue constrained spherical deconvolution (Jeurissen et al., 2014) followed by intensity normalisation (Raffelt et al., 2017). The script `00b_fod_estimation.sh`, shown below, runs these commands across participants.

```

# 1. conversion with gradients embedded
mrconvert data.nii.gz dwi.mif -fslgrad bvecs bvals
mrconvert nodif_brain_mask.nii.gz mask.mif

# 2. tissue response functions (multi-shell)
dwi2response dhollander dwi.mif wm_response.txt gm_response.txt csf_response.txt -mask mask.mif

# 3. group-average response functions (once, across participants)
responsemean */wm_response.txt group_wm_response.txt
responsemean */gm_response.txt group_gm_response.txt
responsemean */csf_response.txt group_csf_response.txt

# 4. multi-shell multi-tissue CSD
dwi2fod msmt_csd dwi.mif group_wm_response.txt wm_fod.mif \
    group_gm_response.txt gm_fod.mif group_csf_response.txt csf_fod.mif -mask mask.mif

# 5. intensity normalisation
mtnormalise wm_fod.mif wm_fod_norm.mif gm_fod.mif gm_fod_norm.mif csf_fod.mif csf_fod_norm.mif
-mask mask.mif

```

Single-shell data may use `dwi2response tournier` and `dwi2fod csd`. The corridor workflow is independent of how the FOD image was estimated.

The script `00b_fod_estimation.sh` is reproduced in the Supplementary Scripts appendix.

FSL route

`probtrackx2` accepts the dilated corridor as a `--waypoints` mask and the exclusions as `--avoid`, with a BEDPOSTX model as input. Its output is voxel-wise connectivity rather than streamlines, so along-tract profiling is not available from it; it is appropriate when a whole-tract summary is sufficient.

Scripts

Each script appears in full on the workflow page for its step. Table S18 lists the set in run order. All scripts read a single configuration file, `00_config.sh`, which is edited once per project. The shell scripts source it themselves; the Python and R scripts read the exported settings, so `source 00_config.sh` is run in the shell before they are called, and again after every edit to the file. Each shell script writes a log to `$OUT/logs/`. Steps 0b, 1, 2, 3, 5, 6, 7 and 8a skip participants whose output already exists (`FORCE=1` recomputes); Steps 4, 8, 8b and 9 recompute on every run. Missing inputs are reported by participant.

Table S18

Scripts, in Run Order

File	Step	Requires
<code>00_config.sh</code>	Project paths, tract definition, parameters, covariate list	bash
<code>00b_fod_estimation.sh</code>	0b. Fibre orientation distributions, when preprocessing ended at the tensor	MRtrix3
<code>01_register_mni_to_t1.sh</code>	1. Registration	ANTs
<code>02_warp_rois.sh</code>	2. Warping of seed, target and atlas	ANTs, FSL
<code>03_build_corridor_mask.sh</code>	3. Corridor and exclusion mask	FSL
<code>04_tune_cutoff.sh</code>	4. Pilot sweep of the cutoff	MRtrix3
<code>04b_compare_cutoffs.py</code>	4. Side-by-side images, Dice overlap, summary table	Python, MRtrix3
<code>05_tractography.sh</code>	5. Tractography	MRtrix3
<code>06_clean_bundles.py</code>	6. Cleaning; streamline count and length of the cleaned bundle	Python, pyAFQ, DIPY
<code>07_visual_qc.py</code>	7. Overlay images and flags	Python, MRtrix3
<code>08a_noddi_fit.py</code>	8. NODDI fit	Python, AMICO
<code>08_node_profiles.py</code>	8. Along-tract profiles	Python, DIPY
<code>08b_build_analysis_csv.py</code>	8. Analysis files for Step 9	Python
<code>09a_tract_models.py</code>	9. Whole-tract and quartile models	Python, statsmodels

File	Step	Requires
09b_nodewise_permutation.R	9. Node-wise permutation test	R
09c_stack_for_explorerer.py	9. Explorer input	Python

Note. NODDI = neurite orientation dispersion and density imaging.

The complete set can be retrieved with the following command.

```
mkdir -p mesoconnect_scripts && cd mesoconnect_scripts
base="https://diffusiontensorimaging-repos.github.io/MesoConnect-Tutorial/scripts"
for f in 00_config.sh 00b_fod_estimation.sh 01_register_mni_to_t1.sh 02_warp_rois.sh \
    03_build_corridor_mask.sh 04_tune_cutoff.sh 04b_compare_cutoffs.py \
    05_tractography.sh 06_clean_bundles.py 07_visual_qc.py 08a_noddi_fit.py \
    08_node_profiles.py 08b_build_analysis_csv.py 09a_tract_models.py \
    09b_nodewise_permutation.R 09c_stack_for_explorerer.py; do
    curl -sSLO "$base/$f"
done
```

Validation status

The scripts generalize those used to process the example dataset, in which every step completed for all 57 participants. The generalized set was run from Step 0b to Step 8b on three participants of that dataset (one tract, left posterior ventral tegmental area → hippocampus), starting from the preprocessed diffusion data and the skull-stripped T1 image, and its outputs were compared with those the example dataset's own scripts had produced for the same participants (Table S19). The group-level scripts, which need a full sample, were checked against the example dataset's stored results (Table S20). The run used MRtrix3 3.0.7, FSL 6.0.5.1, ANTs 2.3.5, Python 3.8.10, DIPY 1.8.0, pyAFQ 1.3.5, AMICO 2.1.1 and bash 5.0 on Linux, with four threads and one participant at a time; it took under an hour.

Table S19

Generalized Scripts Compared With the Example Dataset's Outputs, Three Participants

Output	Agreement
Warped seed, target and atlas (Dice overlap)	.93 to .99
Corridor (Dice overlap)	.98 to .99
Pilot sweep, 0.06 versus 0.01: streamlines generated to reach 1,000	0.39 and 3.0 million versus 0.31 and 0.43 million; Dice between cutoffs .68 and .65 (.66 in the example dataset)
Tractography at 0.01	2,500 streamlines in all three; 0.8 to 1.3 million streamlines generated

Output	Agreement
Cleaned bundle, streamline count	864, 886, 705 versus 857, 824, 733
Cleaned bundle, mean length	within 0.4 mm
Quality-control overlays	No flags; bundles follow the expected arc
NODDI maps	Identical, voxel for voxel
Along-tract profiles (correlation across the 100 nodes)	FA .98; NDI .99 to 1.00; ODI .99; FWF .94 to 1.00

Note. Tractography is stochastic, so streamline counts and profiles are not expected to match exactly. NODDI = neurite orientation dispersion and density imaging; FA = fractional anisotropy; NDI = neurite density index; ODI = orientation dispersion index; FWF = free-water fraction.

Table S20

Checks of the Group-Level Scripts

Script	Check	Result
09a_tract_models.py	Run on the analysis files of the example dataset	Reproduces the reported whole-tract and quartile estimates and interaction tests to three decimals
09b_nodewise_permutation.R	Run on an analysis file of the example dataset	Node-wise <i>t</i> values identical to the stored results; same clusters; identical output with one, three and four cores
09c_stack_for_explorer.py	Run on the outputs of 09b	Explorer file written and loaded
08b_build_analysis_csv.py	Run in the three-participant test	Analysis files written for all four metrics

Two defects surfaced during validation and are fixed in the distributed scripts: on bash 5.0 to 5.2 the shell steps stalled at the end of their first participant loop, and the mean streamline length was read from the wrong line of the `tckstats` output. The scripts were also reviewed independently; the defects found are recorded in the repository history.

The example dataset's own per-step scripts, with its cluster paths and participant list, are in the SDN-IMPACT-DTI repository, Steps 20 to 30 of the ReadMe.

Suggested methods text

The paragraphs below describe the corridor workflow in the form required for a Method section and may be adapted. Values in square brackets are study-specific and should be replaced with the values used; values stated without brackets are those of the workflow as documented here.

The whole tutorial is also available as a single document, with tables and figures numbered S1, S2 and so on and the scripts in an appendix: PDF and Word.

Tractography

Mesolimbic pathways were reconstructed in each participant using the MesoConnect Atlas as an anatomical constraint on participant-level tractography. For each pathway, the seed region, the target region and the atlas's 50% group-overlap map were transformed from Montreal Neurological Institute (MNI) space to each participant's T1-weighted image using symmetric diffeomorphic registration (ANTs; Avants et al., 2008) and then to diffusion space using the affine transform estimated during preprocessing, with nearest-neighbour interpolation. The warped atlas was dilated by [two] voxels and combined with the seed and target regions to define an inclusion zone; its inverse served as the sole exclusion mask. Fibre orientation distributions were estimated with multi-shell multi-tissue constrained spherical deconvolution (Jeurissen et al., 2014) using group-averaged response functions (Dhollander et al., 2016) and intensity normalisation in MRtrix3 (Tournier et al., 2019). Probabilistic tractography (iFOD2; Tournier et al., 2010) was seeded unidirectionally from the seed region, required to enter the target region, and terminated on entry, with [2,500] streamlines selected from a maximum of [25 million] seeds, a fibre orientation distribution amplitude cutoff of [0.01], and length bounds of [35] to [65] mm. The cutoff was selected from a pilot sweep in [five] participants as the most permissive value that reached the streamline target in every pilot participant and reproduced the trajectory obtained at the most conservative cutoff.

Bundle cleaning and quality control

Streamlines were resampled to 100 points and cleaned with pyAFQ (Kruyer et al., 2021; Yeatman et al., 2012) by iteratively removing streamlines more than 3 *SD* (Mahalanobis distance) from the bundle centroid or longer than the mean length by more than 2 *SD*, for up to five iterations. [Where a tract reconstructed as two distinct bundles, streamlines were first clustered with QuickBundles (Garyfallidis et al., 2012; 5 mm threshold), each cluster was cleaned separately, and the anatomically correct cluster was retained after inspection.] Every cleaned bundle was inspected as a tract-density overlay on the mean $b = 0$ image; bundles with fewer than 50 or more than 5,000 tract-density voxels, or a maximum density below 5 streamlines per voxel, were flagged for review. [*n*] participants were excluded at this stage.

Along-tract profiles

Each cleaned bundle was oriented from seed to target against its QuickBundles centroid and sampled at 100 equidistant nodes using Gaussian-weighted tract profiling (Yeatman et al., 2012) as implemented in DIPY (Garyfallidis et al., 2014). [Fractional anisotropy was taken from the diffusion tensor fit. Neurite density index, orientation dispersion index and free-water fraction were obtained from NODDI (Zhang et al., 2012) fitted with AMICO (Daducci et al., 2015), using tissue-weighted modulated maps for neurite density and orientation dispersion (Parker et al., 2021).] Streamline count and mean streamline length were recorded for each bundle as indices of reconstruction quality.

Statistical analysis

[Whole tract: The metric was averaged across nodes and related to the outcome in a linear model (or, for subregional tracts sharing most of their course, a mixed model with a subregion term and a random intercept for participant, in which the outcome $\times$ subregion interaction was tested by likelihood ratio).] [Quartiles: The metric was averaged within nodes 0–24, 25–49, 50–74 and 75–99 and the same model was fitted per quartile, with correction across quartiles by false discovery rate (or maximum-statistic permutation).] [Nodes: At each node the outcome was regressed on the metric and covariates. Adjacent nodes with parametric $p < .05$ formed clusters, and family-wise error was controlled at the cluster level: a null distribution of maximum cluster extent was obtained from 5,000 Freedman–Lane permutations (Freedman & Lane, 1983; Winkler et al., 2014), and clusters were retained when the proportion of permutations with a maximum cluster at least as long did not exceed .05.] Covariates were [intracranial volume, the mean streamline length and streamline count of the cleaned bundle, absolute head motion and age]. [Variables in the whole-tract and quartile models were standardized.] The resolution of analysis and the covariate set were specified before results were examined.

Abbreviations

Abbreviation	Term
ACT	anatomically constrained tractography
AMICO	Accelerated Microstructure Imaging via Convex Optimization
CSD	constrained spherical deconvolution
DIPY	Diffusion Imaging in Python
FA	fractional anisotropy
FDR	false discovery rate
FOD	fibre orientation distribution
FWE	family-wise error
FWF	free-water fraction
HCP	Human Connectome Project
ICV	intracranial volume
iFOD2	second-order integration over fibre orientation distributions
MNI	Montreal Neurological Institute
NDI	neurite density index
NODDI	neurite orientation dispersion and density imaging
ODI	orientation dispersion index
pyAFQ	Python Automated Fiber Quantification
ROI	region of interest
SN	substantia nigra
TDI	tract-density image
VTA	ventral tegmental area

Citation

The atlas manuscript is in preparation. Work that uses the atlas or this workflow should cite that paper; the full reference will be added here when it is published.

Source atlases and software used in the workflow are cited on the pages where they are used and listed on the references page.

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Supplementary scripts

The scripts below are those distributed with the tutorial, in run order.

00_config.sh

```
#!/bin/bash
# =====
# MesoConnect corridor workflow: shared configuration
# =====
# Every step script sources this file. Edit this file only; the step scripts
# contain no project-specific values.
#
# Expected project layout (one directory per participant):
# $PROJECT/anat/<subj>/<subj>_T1w_brain.nii.gz      skull-stripped T1
# $PROJECT/dwi/<subj>/data.nii.gz, bvals, bvecs    preprocessed diffusion data
# $PROJECT/dwi/<subj>/nodif_brain_mask.nii.gz      diffusion-space brain mask
# $PROJECT/dwi/<subj>/mean_b0.nii.gz               background for QC images (optional)
# $PROJECT/dwi/<subj>/fa.nii.gz                   tensor FA map
# $PROJECT/dwi/<subj>/wm_fod_norm.mif              normalized WM FOD (written by 00b)
# $PROJECT/noddi/<subj>/fit_*.nii.gz              NODDI maps (written by 08a)
# $PROJECT/xfm/<subj>/str2diff.mat                 FLIRT T1 -> diffusion affine
#                                                  (omit if T1 and DWI share a grid)
# =====

# --- project -----
export PROJECT="/path/to/project"
export SUBJECTS_FILE="$PROJECT/subjects.txt"      # one participant ID per line
export ATLAS_DIR="/path/to/MesoConnectAtlas"      # atlas and region files, MNI 1 mm
export OUT="$PROJECT/derivatives/mesoconnect"     # everything this workflow writes
export FORCE="${FORCE:-0}"                        # 1 = recompute existing outputs

# --- tract definition (one tract per run; edit and rerun for another tract) ---
export TRACT="l_vta_l_hipp"                      # name used for all outputs
export SEED_MNI="$ATLAS_DIR/roi_maps/left_VTA_0.25_bin.nii.gz"
export TARGET_MNI="$ATLAS_DIR/roi_maps/HPC_L_0.5_bin.nii.gz"
export ATLAS_MNI="$ATLAS_DIR/tracts_thresholded_binary_50/\
l_vta_l_hipp_1mm_MNI_GroupMean_thr50.nii.gz"

# --- T1 -> diffusion transform applied in Step 2 -----
# matrix apply $PROJECT/xfm/<subj>/str2diff.mat (a missing matrix is a missing input)
# header T1 and diffusion images already share a space: resample by image header only
export T1_TO_DWI="matrix"

# --- corridor and tractography parameters (see Reference: Parameters) -----
export DILATE_VOX=2                             # corridor dilation, voxels (1-2; 4 if registration is uncertain)
export CUTOFF=0.01                              # FOD amplitude cutoff selected in Step 4 (MRtrix default 0.05)
export SELECT=2500                              # streamlines to retain
export SEEDS=25000000                           # maximum seeding attempts
export MINLEN=35                                # minimum streamline length, mm
export MAXLEN=65                                # maximum streamline length, mm

# --- group-level models (Steps 8b and 9) -----
export COVARIATES_CSV="$PROJECT/covariates.csv"  # Subject, covariates, outcomes
# Model covariates: columns of $COVARIATES_CSV, plus Mean_length_mm and
# Streamline_count, which Step 6 computes from the cleaned bundle of this tract.
export COVARIATES="ICV,Mean_length_mm,Streamline_count,absolute_motion,age"
export N_PERMUTATIONS=5000
```

```

# --- software and concurrency -----
export MNI_TEMPLATE="$FSLDIR/data/standard/MNI152_T1_1mm_brain.nii.gz"
export ANTSPATH="$ANTSPATH:-/usr/local/ants/bin}"
export PATH="$ANTSPATH:$PATH"
# Peak load is roughly threads x concurrent participants; lower these on a shared machine.
export NTHREADS=8          # threads per MRtrix command
export MAXJOBS=8           # participants processed concurrently in light steps (2, 3)
export ANTS_THREADS=4      # threads per registration (Step 1)
export ANTS_JOBS=4         # registrations run concurrently (Step 1)
export FOD_JOBS=2          # FOD estimations run concurrently (Step 0b; memory-intensive)

# --- helper functions used by the step scripts -----

# Append everything the calling script prints to $OUT/logs/<script>.log.
start_log() {
    mkdir -p "$OUT/logs"
    exec >>(tee -a "$OUT/logs/${basename "$1"}.log") 2>&1
    echo "== $(date '+%F %T')  ${basename "$1"}  TRACT=$TRACT"
}

# Participant IDs as the array SUBJECTS.  Tolerates CRLF line endings, blank lines
# and a missing final newline, none of which a "while read" loop handles.
read_subjects() {
    SUBJECTS=$(tr -d '\r' < "$SUBJECTS_FILE")
    echo "== ${#SUBJECTS[@]} participants in $SUBJECTS_FILE"
}

# Block until fewer than N background jobs are running.  Usage: throttle N
throttle() {
    while [ "$(jobs -r | wc -l)" -ge "$1" ]; do
        sleep 1
    done
}

# Block until every background job has finished.  A bare "wait" must not be used:
# bash 5.0 to 5.2 would also wait for the logging process started by start_log,
# which never exits while the script is running.
wait_for_jobs() {
    throttle 1
}

# Number of non-zero voxels in an image.
nvox() {
    fslstats "$1" -V | awk '{print $1}'
}

# "ok" if the file exists, otherwise "MISSING"; used in the audit tables.
present() {
    if [ -f "$1" ]; then echo ok; else echo MISSING; fi
}

```

01_register_mni_to_t1.sh

```

#!/bin/bash
# =====
# Step 1. Nonlinear registration of the MNI template to each participant's T1
# =====
# Fixed image: participant T1 (skull-stripped).  Moving image: MNI152 1 mm brain.
# The forward transforms (mni2t1_0GenericAffine.mat, mni2t1_1Warp.nii.gz) carry
# MNI-space files into T1 space in Step 2; the inverse warp returns
# participant-level results to MNI space.

```

```

# =====
source "$(dirname "$0")/00_config.sh"
start_log "$0"
read_subjects

register_one() {
    local s=$1
    local t1="$PROJECT/anat/$s/${s}_T1w_brain.nii.gz"
    local d="$OUT/$s/reg"
    if [ ! -f "$t1" ]; then
        echo "!! $s missing $t1"
        return
    fi
    if [ "$FORCE" != 1 ] && [ -f "$d/mni2t1_1Warp.nii.gz" ]; then
        echo "== $s already registered (FORCE=1 to recompute)"
        return
    fi
    mkdir -p "$d"
    antsRegistrationSyNQuick.sh -d 3 \
        -f "$t1" \
        -m "$MNI_TEMPLATE" \
        -o "$d/mni2t1_" \
        -n "$ANTS_THREADS"
    echo ">> $s registered"
}

for s in "${SUBJECTS[@]}; do
    register_one "$s" &
    throttle "$ANTS_JOBS"
done
wait_for_jobs

# Audit
printf "\nSubject\tAffine\tWarp\tInverseWarp\n"
for s in "${SUBJECTS[@]}; do
    d="$OUT/$s/reg"
    printf "%s\t%s\t%s\t%s\n" "$s" \
        "$(present "$d/mni2t1_0GenericAffine.mat")" \
        "$(present "$d/mni2t1_1Warp.nii.gz")" \
        "$(present "$d/mni2t1_1InverseWarp.nii.gz")"
done

```

02_warp_rois.sh

```

#!/bin/bash
# =====
# Step 2. Warp the seed, target and tract atlas into diffusion space
# =====
# MNI -> T1 with the ANTs transforms from Step 1, then T1 -> diffusion with FLIRT,
# using the matrix or the image header according to T1_TO_DWI in 00_config.sh.
# Nearest-neighbour interpolation throughout; outputs are re-binarized.
# =====
source "$(dirname "$0")/00_config.sh"
start_log "$0"
read_subjects

warp_one() {
    local s=$1
    local d="$OUT/$s/rois"
    local t1="$PROJECT/anat/$s/${s}_T1w_brain.nii.gz"
    local ref="$PROJECT/dwi/$s/nodif_brain_mask.nii.gz"

```

```

local warp="$OUT/$s/reg/mni2t1_1Warp.nii.gz"
local aff="$OUT/$s/reg/mni2t1_0GenericAffine.mat"
local mat="$PROJECT/xfm/$s/str2diff.mat"
local name src in_t1 in_diff

if [ "$FORCE" != 1 ] && [ -f "$d/${TRACT}_atlas_diff.nii.gz" ]; then
    echo "== $s already warped"
    return
fi
if [ ! -f "$warp" ] || [ ! -f "$aff" ] || [ ! -f "$ref" ]; then
    echo "!! $s missing registration outputs or diffusion reference"
    return
fi
if [ "$T1_TO_DWI" = matrix ] && [ ! -f "$mat" ]; then
    echo "!! $s missing $mat"
    return
fi
mkdir -p "$d"

for name in seed target atlas; do
    case $name in
        seed)    src=$SEED_MNI ;;
        target) src=$TARGET_MNI ;;
        atlas)  src=$ATLAS_MNI ;;
    esac
    in_t1="$d/${TRACT}_${name}_t1.nii.gz"
    in_diff="$d/${TRACT}_${name}_diff.nii.gz"

    antsApplyTransforms -d 3 -i "$src" -r "$t1" -o "$in_t1" \
        -t "$warp" -t "$aff" -n NearestNeighbor

    if [ "$T1_TO_DWI" = matrix ]; then
        flirt -in "$in_t1" -ref "$ref" -applyxfm -init "$mat" \
            -interp nearestneighbour -out "$in_diff"
    else
        # T1 and diffusion share a space: resample onto the diffusion grid by header
        flirt -in "$in_t1" -ref "$ref" -applyxfm -usesqform \
            -interp nearestneighbour -out "$in_diff"
    fi
    fslmaths "$in_diff" -thr 0.5 -bin "$in_diff"
done
echo ">> $s warped"
}

case "$T1_TO_DWI" in
    matrix) echo "== T1 -> diffusion: applying xfm/<sub>j>/str2diff.mat" ;;
    header) echo "== T1 -> diffusion: resampling by image header (no matrix)" ;;
    *)      echo "!! T1_TO_DWI must be matrix or header"; exit 1 ;;
esac

for s in "${SUBJECTS[@]"; do
    warp_one "$s" &
    throttle "$MAXJOBS"
done
wait_for_jobs

# Audit: voxel counts and seed-target overlap (the overlap must be 0)
tmp=$(mktemp -d)
printf "\nSubject\tseed_vox\ttarget_vox\tatlas_vox\tseed_target_overlap\n"
for s in "${SUBJECTS[@]"; do
    d="$OUT/$s/rois"
    if [ ! -f "$d/${TRACT}_atlas_diff.nii.gz" ]; then
        printf "%s\tMISSING\n" "$s"
    fi

```

```

        continue
    fi
    fslmaths "$d/${TRACT}_seed_diff.nii.gz" -mul "$d/${TRACT}_target_diff.nii.gz" "$tmp/overlap"
    printf "%s\t%s\t%s\t%s\t%s\n" "$s" \
        "$(nvox "$d/${TRACT}_seed_diff.nii.gz")" \
        "$(nvox "$d/${TRACT}_target_diff.nii.gz")" \
        "$(nvox "$d/${TRACT}_atlas_diff.nii.gz")" \
        "$(nvox "$tmp/overlap")"
done
rm -rf "$tmp"

```

03_build_corridor_mask.sh

```

#!/bin/bash
# =====
# Step 3. Build the corridor and its exclusion mask
# =====
# Dilate the warped atlas, add the seed and target, binarize (inclusion zone),
# then invert. The inverted image is the single exclusion mask given to tckgen:
# every voxel outside the corridor terminates and discards a streamline.
# =====
source "$(dirname "$0")/00_config.sh"
start_log "$0"
read_subjects

# One -dilm pass (3 x 3 x 3 kernel) per voxel of requested dilation
DILATE_ARGS=""
for ((i = 0; i < DILATE_VOX; i++)); do
    DILATE_ARGS="$DILATE_ARGS -dilm"
done

build_one() {
    local s=$1
    local d="$OUT/$s/rois"
    if [ "$FORCE" != 1 ] && [ -f "$d/${TRACT}_exclusion_mask.nii.gz" ]; then
        echo "== $s corridor exists"
        return
    fi
    if [ ! -f "$d/${TRACT}_atlas_diff.nii.gz" ]; then
        echo "!! $s missing warped atlas (run Step 2)"
        return
    fi
    # $DILATE_ARGS is intentionally unquoted so that it expands to separate options
    fslmaths "$d/${TRACT}_atlas_diff.nii.gz" $DILATE_ARGS "$d/${TRACT}_atlas_dilated.nii.gz"
    fslmaths "$d/${TRACT}_atlas_dilated.nii.gz" \
        -add "$d/${TRACT}_seed_diff.nii.gz" \
        -add "$d/${TRACT}_target_diff.nii.gz" \
        -bin "$d/${TRACT}_inclusion_zone.nii.gz"
    fslmaths "$d/${TRACT}_inclusion_zone.nii.gz" -binv "$d/${TRACT}_exclusion_mask.nii.gz"
    echo ">> $s corridor: $(nvox "$d/${TRACT}_inclusion_zone.nii.gz") voxels"
}

for s in "${SUBJECTS[@]}; do
    build_one "$s" &
    throttle "$MAXJOBS"
done
wait_for_jobs

# Audit: no seed or target voxel may fall inside the exclusion mask (both counts 0)
tmp=$(mktemp -d)
printf "\nSubject\tseed_excluded\ttarget_excluded\tcorridor_vox\n"

```

```

for s in "${SUBJECTS[@]}"; do
    d="$OUT/$s/rois"
    excl="$d/${TRACT}_exclusion_mask.nii.gz"
    if [ ! -f "$excl" ]; then
        printf "%s\tMISSING\n" "$s"
        continue
    fi
    fslmaths "$excl" -mul "$d/${TRACT}_seed_diff.nii.gz" "$tmp/seed"
    fslmaths "$excl" -mul "$d/${TRACT}_target_diff.nii.gz" "$tmp/target"
    printf "%s\t%s\t%s\t%s\n" "$s" \
        "$(nvox "$tmp/seed")" \
        "$(nvox "$tmp/target")" \
        "$(nvox "$d/${TRACT}_inclusion_zone.nii.gz")"
done
rm -rf "$tmp"

```

04_tune_cutoff.sh

```

#!/bin/bash
# =====
# Step 4. Pilot sweep of the FOD amplitude cutoff
# =====
# Runs reduced-budget tractography at several cutoffs in a few participants.
# "Selected" is the number of streamlines that met every criterion; "Generated" is
# the number tckgen had to generate (selected plus rejected) to obtain them. tckgen
# generates one streamline per seed, so it is also the number of seeds consumed.
# Usage:
#   bash 04_tune_cutoff.sh "sub-01 sub-02 sub-03 sub-04 sub-05" "0.1 0.08 0.06 0.01"
# Defaults: the first five participants in $SUBJECTS_FILE and the four cutoffs above.
# Follow with 04b_compare_cutoffs.py for the side-by-side images and summary table.
# =====
source "$(dirname "$0")/00_config.sh"
start_log "$0"
read_subjects

PILOT=${1:-${SUBJECTS[*]:0:5}}
CUTOFFS=${2:-"0.1 0.08 0.06 0.01"}
PILOT_SELECT=1000
PILOT_SEEDS=5000000

printf "%-12s %-8s %-12s %-12s %-12s\n" Subject Cutoff Selected Generated MeanLen_mm
for s in $PILOT; do
    rois="$OUT/$s/rois"
    tdir="$OUT/$s/tckgen/${TRACT}"
    if [ ! -f "$rois/${TRACT}_exclusion_mask.nii.gz" ]; then
        echo "!! $s missing corridor (run Steps 2 and 3)"
        continue
    fi
    mkdir -p "$tdir"
    for c in $CUTOFFS; do
        tck="$tdir/${TRACT}_pilot_${c}.tck"
        tckgen "$PROJECT/dwi/$s/wm_fod_norm.mif" "$tck" \
            -algorithm iFOD2 \
            -seed_image "$rois/${TRACT}_seed_diff.nii.gz" -seed_unidirectional \
            -include "$rois/${TRACT}_target_diff.nii.gz" \
            -exclude "$rois/${TRACT}_exclusion_mask.nii.gz" \
            -select "$PILOT_SELECT" -seeds "$PILOT_SEEDS" -cutoff "$c" \
            -minlength "$MINLEN" -maxlength "$MAXLEN" -stop \
            -nthreads "$NTHREADS" -force -quiet
        count=$(tckinfo "$tck" | awk '$1 == "count:" {print $2}')
        generated=$(tckinfo "$tck" | awk '$1 == "total_count:" {print $2}')
    done
done

```

```

    if [ "${count:-0}" -gt 0 ]; then
        meanlen=$(tckstats "$tck" -output mean -quiet | awk '{print $1}')
    else
        meanlen=NA
    fi
    printf "%-12s %-8s %-12s %-12s %-12s\n" "$s" "$c" "$count" "$generated" "$meanlen"
done
done
echo "Reaching the streamline target in every pilot participant is necessary but not"
echo "sufficient: compare the reconstructions with 04b_compare_cutoffs.py before choosing."

```

04b_compare_cutoffs.py

```

#!/usr/bin/env python3
"""Step 4b. Comparison of the pilot cutoffs.

```

Run after 04_tune_cutoff.sh, in a shell where 00_config.sh has been sourced:

```

source 00_config.sh
python 04b_compare_cutoffs.py "sub-01 sub-02 sub-03" "0.1 0.08 0.06 0.01"

```

For every pilot participant the script renders the tract-density image of each cutoff side by side (axial row, coronal row). It then writes cutoff_summary.csv (streamlines selected, streamlines generated, occupied voxels, Dice overlap with the most conservative cutoff, mean and SD of streamline length) and cutoff_summary.png.

```

Outputs: $OUT/qc/<TRACT>_cutoff_pilot/
"""

```

```

import os
import subprocess
import sys
from pathlib import Path

```

```

import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
import nibabel as nib
import numpy as np
import pandas as pd

```

```

def env(name):
    if name not in os.environ:
        sys.exit(f"{name} is not set: run `source 00_config.sh` first")
    return os.environ[name]

```

```

PROJECT = Path(env("PROJECT"))
OUT = Path(env("OUT"))
TRACT = env("TRACT")

```

```

if len(sys.argv) > 1:
    PILOT = sys.argv[1].split()
else:
    PILOT = Path(env("SUBJECTS_FILE")).read_text().split()[:5]
CUTOFFS = sys.argv[2].split() if len(sys.argv) > 2 else ["0.1", "0.08", "0.06", "0.01"]
REFERENCE = CUTOFFS[0] # most conservative cutoff: reference for Dice overlap

```

```

QC = OUT / "qc" / f"{TRACT}_cutoff_pilot"
QC.mkdir(parents=True, exist_ok=True)

```

```

def run(cmd):
    return subprocess.run(cmd, capture_output=True, text=True, check=True).stdout

def tck_summary(tck):
    """Streamlines selected, streamlines generated (selected plus rejected), and the
    mean and SD of length (mm)."""
    info = {}
    for line in run(["tckinfo", str(tck)]).splitlines():
        key, _, value = line.strip().partition(":")
        info[key] = value.strip()
    count = int(info.get("count", 0))
    generated = int(info.get("total_count", 0))
    if count == 0:
        return count, generated, np.nan, np.nan
    mean = float(run(["tckstats", str(tck), "-output", "mean", "-quiet"]))
    sd = float(run(["tckstats", str(tck), "-output", "std", "-quiet"]))
    return count, generated, mean, sd

def density_mask(tck, template, scratch):
    """Binary mask of the voxels visited by at least one streamline."""
    run(["tckmap", str(tck), str(scratch), "-template", str(template), "-force", "-quiet"])
    mask = nib.load(str(scratch)).get_fdata() > 0
    scratch.unlink()
    return mask

def dice(a, b):
    total = a.sum() + b.sum()
    return 2 * np.logical_and(a, b).sum() / total if total else np.nan

def overlay(ax, background, mask, title):
    ax.imshow(np.rot90(background), cmap="gray")
    ax.imshow(np.rot90(np.ma.masked_where(~mask, mask.astype(float))),
              cmap="autumn", alpha=0.8, vmin=0, vmax=1)
    ax.set_title(title, fontsize=10)
    ax.axis("off")

rows = []
for s in PILOT:
    template = PROJECT / "dwi" / s / "nodif_brain_mask.nii.gz"
    b0 = PROJECT / "dwi" / s / "mean_b0.nii.gz"
    background = nib.load(str(b0 if b0.exists() else template)).get_fdata()

    masks = {}
    for c in CUTOFFS:
        tck = OUT / s / "tckgen" / TRACT / f"{TRACT}_pilot_{c}.tck"
        if not tck.exists():
            print(f"[{s}] cutoff {c}: no tractogram (run 04_tune_cutoff.sh)")
            continue
        count, generated, mean, sd = tck_summary(tck)
        masks[c] = density_mask(tck, template, QC / f"_{s}_{c}_tdi.nii.gz")
        rows.append({
            "subject": s, "cutoff": c, "selected": count, "generated": generated,
            "voxels": int(masks[c].sum()), "mean_len_mm": mean, "sd_len_mm": sd,
        })
    if not masks:
        continue

```

```

# Dice overlap of every cutoff with the reference cutoff
for row in rows:
    if row["subject"] == s and REFERENCE in masks:
        row["dice_vs_reference"] = round(dice(masks[REFERENCE], masks[row["cutoff"]]), 3)

# Slices through the densest part of the union of all cutoffs
union = np.logical_or.reduce(list(masks.values()))
k_axial = int(np.argmax(union.sum(axis=(0, 1))))
k_coronal = int(np.argmax(union.sum(axis=(0, 2))))

fig, axes = plt.subplots(2, len(masks), figsize=(3.2 * len(masks), 6.4), squeeze=False)
for j, (c, mask) in enumerate(masks.items()):
    overlay(axes[0][j], background[:, :, k_axial], mask[:, :, k_axial],
            f"Axial, cutoff {c}")
    overlay(axes[1][j], background[:, k_coronal, :], mask[:, k_coronal, :],
            f"Coronal, cutoff {c}")
fig.suptitle(f"{s} {TRACT} pilot cutoffs (uncleaned)", fontsize=11)
fig.tight_layout()
fig.savefig(QC / f"{s}_cutoff_compare.png", dpi=110, bbox_inches="tight")
plt.close(fig)
print(f"[{s}] wrote {s}_cutoff_compare.png")

if not rows:
    sys.exit("no pilot tractograms found")

summary = pd.DataFrame(rows)
summary.to_csv(QC / "cutoff_summary.csv", index=False)

# Mean across pilot participants, one bar per cutoff
panels = [("selected", "Streamlines selected"), ("generated", "Streamlines generated"),
          ("voxels", "Occupied voxels"), ("sd_len_mm", "SD of length (mm)")]
means = summary.groupby("cutoff", sort=False)[[k for k, _ in panels]].mean()
fig, axes = plt.subplots(1, len(panels), figsize=(15, 3.6))
for ax, (key, title) in zip(axes, panels):
    ax.bar(means.index.astype(str), means[key], color="#3A6B8C")
    ax.set_title(title)
    ax.set_xlabel("FOD amplitude cutoff")
fig.tight_layout()
fig.savefig(QC / "cutoff_summary.png", dpi=110)
plt.close(fig)

print(summary.to_string(index=False))
print(f"\nsummary -> {QC / 'cutoff_summary.csv'} and cutoff_summary.png")

```

05_tractography.sh

```

#!/bin/bash
# =====
# Step 5. Corridor-constrained tractography for every participant
# =====
# Seeds in the seed region, requires the target region, and discards any
# streamline that leaves the corridor (single exclusion mask from Step 3).
# Participants are processed serially; tckgen uses $NTHREADS threads.
# Writes <tract>_<cutoff>.tck per participant and a summary of streamlines
# selected and streamlines generated (selected plus rejected). The streamline
# count and length used as model covariates are taken from the cleaned bundles
# in Step 6, not from this step.
# =====
source "$(dirname "$0")/00_config.sh"
start_log "$0"
read_subjects

```

```

for s in "${SUBJECTS[@]}"; do
    rois="$OUT/$s/rois"
    tdir="$OUT/$s/tckgen/$TRACT"
    tck="$tdir/${TRACT}_${CUTOFF}.tck"
    if [ ! -f "$PROJECT/dwi/$s/wm_fod_norm.mif" ]; then
        echo "!! $s missing FOD image (run 00b)"
        continue
    fi
    if [ ! -f "$rois/${TRACT}_exclusion_mask.nii.gz" ]; then
        echo "!! $s missing corridor (run Steps 2 and 3)"
        continue
    fi
    if [ "$FORCE" != 1 ] && [ -f "$tck" ]; then
        echo "== $s tractogram exists"
        continue
    fi
    mkdir -p "$tdir"
    # tckgen creates its output file at start-up; writing to a temporary name keeps an
    # interrupted run from being mistaken for a finished tractogram on restart.
    partial="$tdir/${TRACT}_${CUTOFF}.partial.tck"
    if tckgen "$PROJECT/dwi/$s/wm_fod_norm.mif" "$partial" \
        -algorithm iFOD2 \
        -seed_image "$rois/${TRACT}_seed_diff.nii.gz" -seed_unidirectional \
        -include "$rois/${TRACT}_target_diff.nii.gz" \
        -exclude "$rois/${TRACT}_exclusion_mask.nii.gz" \
        -select "$SELECT" -seeds "$SEEDS" -cutoff "$CUTOFF" \
        -minlength "$MINLEN" -maxlength "$MAXLEN" -stop \
        -nthreads "$NTHREADS" -force; then
        mv -f "$partial" "$tck"
        echo ">> $s finished"
    else
        rm -f "$partial"
        echo "!! $s tckgen failed or was interrupted"
    fi
done

# Audit and summary table (rebuilt from the tractograms on every run)
mkdir -p "$OUT/qc"
stats="$OUT/qc/${TRACT}_tckgen_summary.csv"
echo "Subject,Selected,MeanLength_mm,Generated" > "$stats"
reached=0
total=0
for s in "${SUBJECTS[@]}"; do
    tck="$OUT/$s/tckgen/$TRACT/${TRACT}_${CUTOFF}.tck"
    total=$((total + 1))
    if [ ! -f "$tck" ]; then
        echo "MISSING: $s"
        continue
    fi
    count=$(tckinfo "$tck" | awk ' $1 == "count:" {print $2}')
    generated=$(tckinfo "$tck" | awk ' $1 == "total_count:" {print $2}')
    if [ "$count:-0" -gt 0 ]; then
        meanlen=$(tckstats "$tck" -output mean -quiet | awk '{print $1}')
    else
        meanlen=NA
    fi
    echo "$s,$count,$meanlen,$generated" >> "$stats"
    if [ "$count:-0" -ge "$SELECT" ]; then
        reached=$((reached + 1))
    else
        echo "SHORT: $s ($count selected of $generated generated)"
    fi
done

```

```
done
echo "reached $SELECT streamlines: $reached / $total"
echo "summary -> $stats"
```

06_clean_bundles.py

```
#!/usr/bin/env python3
"""Step 6. Bundle cleaning (pyAFQ) and the cleaned-bundle covariate table.
```

Run in a shell where 00_config.sh has been sourced:

```
source 00_config.sh
python 06_clean_bundles.py
```

Each tractogram from Step 5 is cleaned with `AFQ.recognition.cleaning.clean_bundle`: streamlines are resampled to 100 points, and those farther than `DISTANCE_SD` from the bundle core (Mahalanobis distance), or longer than the mean length by more than `LENGTH_SD`, are removed, for up to `CLEAN_ROUNDS` iterations.

Some tracts reconstruct as two distinct bundles in a subset of participants. With `QB_SPLIT = True` the tractogram is first divided with `QuickBundles`, the two largest clusters are cleaned separately, and the anatomically correct cluster is chosen by inspection and copied to `<TRACT>_<CUTOFF>_cleaned.tck`. Rerun the script afterwards to rebuild the covariate table.

Outputs

```
$OUT/<subj>/tckgen/<TRACT>/<TRACT>_<CUTOFF>_cleaned.tck
$OUT/nodewise/<TRACT>_tract_stats.csv
    Subject, Streamline_count, Mean_length_mm (cleaned bundle; model covariates),
    Count_uncleaned, Retention_pct
```

```
"""
```

```
import inspect
import os
import sys
from pathlib import Path
```

```
import pandas as pd
try:
    from AFQ.recognition.cleaning import clean_bundle
except ImportError:
    # pyAFQ 1.3.2 and earlier
    from AFQ.segmentation import clean_bundle
from dipy.io.stateful_tractogram import StatefulTractogram
from dipy.io.streamline import load_tractogram, save_tractogram
from dipy.segment.clustering import QuickBundles
from dipy.segment.featurespeed import ResampleFeature
from dipy.segment.metricspeed import AveragePointwiseEuclideanMetric
from dipy.tracking.streamline import length
```

```
def env(name):
    if name not in os.environ:
        sys.exit(f"{name} is not set: run `source 00_config.sh` first")
    return os.environ[name]
```

```
PROJECT = Path(env("PROJECT"))
OUT = Path(env("OUT"))
TRACT = env("TRACT")
CUTOFF = env("CUTOFF")
FORCE = os.environ.get("FORCE", "0") == "1"
SUBJECTS = Path(env("SUBJECTS_FILE")).read_text().split()
```

```

N_POINTS = 100          # resampling for the Mahalanobis distance
CLEAN_ROUNDS = 5         # maximum cleaning iterations
DISTANCE_SD = 3          # Mahalanobis distance threshold, SD
LENGTH_SD = 2           # length threshold, SD above the mean
QB_SPLIT = False        # True: split into two bundles before cleaning
QB_THRESHOLD_MM = 5.0    # QuickBundles distance threshold

CLEAN_ARGS = dict(n_points=N_POINTS, clean_rounds=CLEAN_ROUNDS, stat="mean",
                  distance_threshold=DISTANCE_SD, length_threshold=LENGTH_SD)
# pyAFQ 3 and later compute the distance over the middle 60% of each streamline by
# default; core_only=0 restores the full-length computation of earlier versions.
if "core_only" in inspect.signature(clean_bundle).parameters:
    CLEAN_ARGS["core_only"] = 0

def clean(sft):
    # The cleaned bundle is rebuilt from the retained indices, so that its coordinate
    # space is that of the input under every pyAFQ version.
    _, keep = clean_bundle(sft, return_idx=True, **CLEAN_ARGS)
    return StatefulTractogram.from_sft(sft.streamlines[keep], sft)

def load(tck, reference):
    return load_tractogram(str(tck), str(reference), bbox_valid_check=False)

for s in SUBJECTS:
    tdir = OUT / s / "tckgen" / TRACT
    raw = tdir / f"{TRACT}_{CUTOFF}.tck"
    out = tdir / f"{TRACT}_{CUTOFF}_cleaned.tck"
    reference = PROJECT / "dwi" / s / "nodif_brain_mask.nii.gz"

    if not raw.exists():
        print(f"[{s}] SKIP: no tractogram from Step 5")
        continue
    if out.exists() and not FORCE:
        print(f"[{s}] cleaned bundle exists")
        continue
    sft = load(raw, reference)
    n_raw = len(sft.streamlines)
    if n_raw == 0:
        print(f"[{s}] SKIP: 0 streamlines")
        continue

    if not QB_SPLIT:
        cleaned = clean(sft)
        save_tractogram(cleaned, str(out), bbox_valid_check=False)
        n_clean = len(cleaned.streamlines)
        print(f"[{s}] {n_raw} -> {n_clean} streamlines ({100 * n_clean / n_raw:.0f}% retained)")
        continue

    metric = AveragePointwiseEuclideanMetric(ResampleFeature(nb_points=N_POINTS))
    clusters = QuickBundles(threshold=QB_THRESHOLD_MM, metric=metric).cluster(sft.streamlines)
    largest = sorted(clusters, key=len, reverse=True)[:2]
    for i, cluster in enumerate(largest, start=1):
        part = StatefulTractogram.from_sft(sft.streamlines[cluster.indices], sft)
        cleaned = clean(part)
        name = f"{TRACT}_{CUTOFF}_cluster{i}_cleaned.tck"
        save_tractogram(cleaned, str(tdir / name), bbox_valid_check=False)
        print(f"[{s}] cluster {i}: {len(cluster)} -> {len(cleaned.streamlines)} streamlines")
    print(f"[{s}] inspect both clusters; copy the correct one to {out.name}")

```

```

# Covariate table from the cleaned bundles (rebuilt on every run)
rows = []
for s in SUBJECTS:
    tdir = OUT / s / "tckgen" / TRACT
    raw = tdir / f"{TRACT}_{CUTOFF}.tck"
    out = tdir / f"{TRACT}_{CUTOFF}_cleaned.tck"
    if not out.exists():
        continue
    reference = PROJECT / "dwi" / s / "nodif_brain_mask.nii.gz"
    lengths = length(load(out, reference).streamlines)
    n_raw = len(load(raw, reference).streamlines) if raw.exists() else float("nan")
    rows.append({
        "Subject": s,
        "Streamline_count": len(lengths),
        "Mean_length_mm": round(float(lengths.mean()), 3) if len(lengths) else float("nan"),
        "Count_uncleaned": n_raw,
        "Retention_pct": round(100 * len(lengths) / n_raw, 1) if n_raw else float("nan"),
    })

table = OUT / "nodewise" / f"{TRACT}_tract_stats.csv"
table.parent.mkdir(parents=True, exist_ok=True)
pd.DataFrame(rows).to_csv(table, index=False)
print(f"\n{len(rows)} of {len(SUBJECTS)} participants have a cleaned bundle")
print(f"covariate table -> {table}")

```

07_visual_qc.py

```

#!/usr/bin/env python3
"""Step 7. Tract-density overlays of every cleaned bundle, with automatic flags.

```

Run in a shell where 00_config.sh has been sourced:

```

source 00_config.sh
python 07_visual_qc.py

```

For each participant the cleaned bundle is converted to a tract-density image (tckmap) and drawn over the mean $b = 0$ image in the axial and coronal slices that contain the most streamlines. Flags mark bundles that need inspection in a viewer; they do not replace review of the unflagged images.

Outputs: \$OUT/qc/<TRACT>/<subj>_<TRACT>_qc.png and qc_flags.csv

```

"""
import os
import subprocess
import sys
from pathlib import Path

import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
import nibabel as nib
import numpy as np
import pandas as pd

def env(name):
    if name not in os.environ:
        sys.exit(f"{name} is not set: run `source 00_config.sh` first")
    return os.environ[name]

```

```

PROJECT = Path(env("PROJECT"))
OUT = Path(env("OUT"))
TRACT = env("TRACT")
CUTOFF = env("CUTOFF")
FORCE = os.environ.get("FORCE", "0") == "1"
SUBJECTS = Path(env("SUBJECTS_FILE")).read_text().split()

MIN_VOXELS = 50          # fewer occupied voxels: LOW_COVERAGE
MAX_VOXELS = 5000        # more occupied voxels: HIGH_COVERAGE (bundle leaves the pathway)
MIN_PEAK_DENSITY = 5     # peak streamlines per voxel below this: LOW_DENSITY

QC = OUT / "qc" / TRACT
QC.mkdir(parents=True, exist_ok=True)

def flag_for(n_voxels, peak):
    if n_voxels < MIN_VOXELS:
        return "LOW_COVERAGE"
    if n_voxels > MAX_VOXELS:
        return "HIGH_COVERAGE"
    if peak < MIN_PEAK_DENSITY:
        return "LOW_DENSITY"
    return ""

def overlay(ax, background, density, title):
    ax.imshow(np.rot90(background), cmap="gray")
    ax.imshow(np.rot90(np.ma.masked_where(density <= 0, density)), cmap="hot", alpha=0.7)
    ax.set_title(title)
    ax.axis("off")

rows = []
for s in SUBJECTS:
    tck = OUT / s / "tckgen" / TRACT / f"{TRACT}_{CUTOFF}_cleaned.tck"
    template = PROJECT / "dwi" / s / "nodif_brain_mask.nii.gz"
    b0 = PROJECT / "dwi" / s / "mean_b0.nii.gz"
    png = QC / f"{s}_{TRACT}_qc.png"

    if not tck.exists():
        rows.append({"Subject": s, "Voxels": 0, "Peak_density": 0, "Flag": "MISSING"})
        print(f"[{s}] MISSING cleaned bundle")
        continue

    tdi = QC / f"{s}_tdi.nii.gz"
    subprocess.run(["tckmap", str(tck), str(tdi), "-template", str(template),
                    "-force", "-quiet"], check=True)
    density = nib.load(str(tdi)).get_fdata()
    tdi.unlink()

    n_voxels = int((density > 0).sum())
    peak = float(density.max())
    flag = flag_for(n_voxels, peak)
    rows.append({"Subject": s, "Voxels": n_voxels, "Peak_density": peak, "Flag": flag})
    print(f"[{s}] {n_voxels} voxels, peak {peak:.0f} {flag}")

    if png.exists() and not FORCE:
        continue

    background = nib.load(str(b0 if b0.exists() else template)).get_fdata()
    k_axial = int(np.argmax(density.sum(axis=(0, 1))))
    k_coronal = int(np.argmax(density.sum(axis=(0, 2))))
    fig, axes = plt.subplots(1, 2, figsize=(10, 5))

```

```

overlay(axes[0], background[:, :, k_axial], density[:, :, k_axial], "Axial")
overlay(axes[1], background[:, k_coronal, :], density[:, k_coronal, :], "Coronal")
fig.suptitle(f"{s} {TRACT} cleaned" + (f" [{flag}]" if flag else ""), fontsize=11)
fig.tight_layout()
fig.savefig(png, dpi=100, bbox_inches="tight")
plt.close(fig)

flags = pd.DataFrame(rows)
flags.to_csv(QC / "qc_flags.csv", index=False)
flagged = flags[flags["Flag"] != ""]
print(f"\n{len(flagged)} of {len(flags)} participants flagged")
if len(flagged):
    print(flagged.to_string(index=False))
print(f"flags -> {QC / 'qc_flags.csv'}")

```

08_node_profiles.py

```

#!/usr/bin/env python3
"""Step 8. Along-tract profiles at 100 nodes (AFQ, Gaussian-weighted).

```

Run in a shell where 00_config.sh has been sourced:

```

source 00_config.sh
python 08_node_profiles.py

```

Streamlines are first oriented to the bundle centroid and the centroid is oriented so that it starts at the seed region; node 0 is therefore the seed end and node 99 the target end in every participant. At each node, every streamline's value is weighted by the inverse of its Mahalanobis distance from the bundle core (Yeatman et al., 2012).

METRICS maps a column name to a scalar image; add or remove entries freely. A metric whose image exists for no participant (for example NODDI when Step 8a was not run) is dropped with a notice; a participant missing any remaining image is skipped.

```

Output: $OUT/nodewise/<TRACT>_nodewise_all_subjects.csv
        (long format: Subject, Tract, Node, one column per metric)
"""
import os
import sys
from pathlib import Path

import nibabel as nib
import numpy as np
import pandas as pd
from dipy.io.image import load_nifti
from dipy.io.streamline import load_tractogram
from dipy.segment.clustering import QuickBundles
from dipy.segment.featurespeed import ResampleFeature
from dipy.segment.metricspeed import AveragePointwiseEuclideanMetric
from dipy.stats.analysis import afq_profile, gaussian_weights
from dipy.tracking.streamline import Streamlines, orient_by_streamline

def env(name):
    if name not in os.environ:
        sys.exit(f"{name} is not set: run `source 00_config.sh` first")
    return os.environ[name]

PROJECT = Path(env("PROJECT"))

```

```

OUT = Path(env("OUT"))
TRACT = env("TRACT")
CUTOFF = env("CUTOFF")
SUBJECTS = Path(env("SUBJECTS_FILE")).read_text().split()

N_NODES = 100
MIN_STREAMLINES = 5
METRICS = {
    "FA": "dwi/{s}/fa.nii.gz",
    "NDI": "noddif/{s}/fit_NDI_modulated.nii.gz",
    "ODI": "noddif/{s}/fit_ODI_modulated.nii.gz",
    "FWF": "noddif/{s}/fit_FWF.nii.gz",
}

unavailable = [m for m, template in METRICS.items()
                if not any((PROJECT / template.format(s=s)).exists() for s in SUBJECTS)]
for m in unavailable:
    print(f"metric {m}: no image found for any participant; not profiled")
    del METRICS[m]
if not METRICS:
    sys.exit("no scalar images found")

def seed_centre_mm(s):
    """Centre of mass of the warped seed region, in scanner (mm) coordinates."""
    img = nib.load(str(OUT / s / "rois" / f"{TRACT}_seed_diff.nii.gz"))
    voxels = np.argwhere(img.get_fdata() > 0)
    return nib.affines.apply_affine(img.affine, voxels.mean(axis=0))

def orient_seed_to_target(streamlines, seed_mm):
    """Orient all streamlines alike, with node 0 at the seed end."""
    metric = AveragePointwiseEuclideanMetric(ResampleFeature(nb_points=N_NODES))
    centroid = QuickBundles(threshold=np.inf, metric=metric).cluster(streamlines).centroids[0]
    starts_at_target = (np.linalg.norm(centroid[0] - seed_mm)
                        > np.linalg.norm(centroid[-1] - seed_mm))
    if starts_at_target:
        centroid = centroid[::-1]
    return Streamlines(orient_by_streamline(streamlines, centroid))

def profile(image, streamlines, weights):
    data, affine = load_nifti(str(image))
    return afq_profile(data, streamlines, affine, n_points=N_NODES, weights=weights)

rows = []
for s in SUBJECTS:
    tck = OUT / s / "tckgen" / TRACT / f"{TRACT}_{CUTOFF}_cleaned.tck"
    maps = {m: PROJECT / template.format(s=s) for m, template in METRICS.items()}
    missing = [str(p) for p in [tck, *maps.values()] if not p.exists()]
    if missing:
        print(f"[{s}] SKIP: missing {missing[0]}")
        continue

    reference = next(iter(maps.values()))
    streamlines = load_tractogram(str(tck), str(reference), bbox_valid_check=False).streamlines
    if len(streamlines) < MIN_STREAMLINES:
        print(f"[{s}] SKIP: {len(streamlines)} streamlines")
        continue

    streamlines = orient_seed_to_target(streamlines, seed_centre_mm(s))

```

```

weights = gaussian_weights(streamlines, n_points=N_NODES)
profiles = {m: profile(p, streamlines, weights) for m, p in maps.items()}
for node in range(N_NODES):
    rows.append({"Subject": s, "Tract": TRACT, "Node": node,
                **{m: float(profiles[m][node]) for m in METRICS}})
print(f"[{s}] {len(streamlines)} streamlines, {N_NODES} nodes, {len(METRICS)} metrics")

out = OUT / "nodewise" / f"{TRACT}_nodewise_all_subjects.csv"
out.parent.mkdir(parents=True, exist_ok=True)
pd.DataFrame(rows).to_csv(out, index=False)
print(f"DONE -> {out} ({len(rows) // N_NODES} participants)")

```

08a_noddi_fit.py

```

#!/usr/bin/env python3
"""Step 8a. NODDI fit with AMICO (run before Step 8 when NODDI metrics are profiled).

```

Run in a shell where 00_config.sh has been sourced:

```

source 00_config.sh
python 08a_noddi_fit.py                # every participant in $SUBJECTS_FILE
python 08a_noddi_fit.py sub-01 sub-02  # selected participants

```

Writes fit_NDI, fit_ODI and fit_FWF, the tissue-weighted maps fit_NDI_modulated and fit_ODI_modulated (Parker et al., 2021), and fit_RMSE. Tract profiles use the modulated NDI and ODI maps and the unmodulated FWF map.

The intrinsic parallel diffusivity defaults to the white-matter value (1.7e-3 mm²/s). For gray-matter regions set NODDI_DPAR=1.1e-3; results are then written to \$PROJECT/noddi_gm so that white-matter maps are not overwritten. Response kernels are generated per participant (work/<subj>/kernels), so several instances of the script can run concurrently on different participants.

Outputs: \$PROJECT/noddi/<subj>/ (or \$PROJECT/noddi_gm/<subj>/)

```

"""
import os
import sys
from pathlib import Path

N_THREADS = os.environ.get("NODDI_NTHREADS", "4")
os.environ["OPENBLAS_NUM_THREADS"] = N_THREADS      # must be set before amico is imported
os.environ["OMP_NUM_THREADS"] = N_THREADS

import amico

def env(name):
    if name not in os.environ:
        sys.exit(f"{name} is not set: run `source 00_config.sh` first")
    return os.environ[name]

PROJECT = Path(env("PROJECT"))
FORCE = os.environ.get("FORCE", "0") == "1"
SUBJECTS = sys.argv[1:] or Path(env("SUBJECTS_FILE")).read_text().split()

D_PAR = os.environ.get("NODDI_DPAR")      # mm2/s; unset = AMICO default for white matter
D_ISO = 3.0e-3                            # isotropic (free water) diffusivity, mm2/s
B0_THRESHOLD = 100                        # volumes with b below this are treated as b = 0
B_STEP = 200                             # b-values are rounded to the nearest multiple

```

```

STUDY = PROJECT / ("noddi_gm" if D_PAR else "noddi")
STUDY.mkdir(parents=True, exist_ok=True)
amico.setup()

for s in SUBJECTS:
    dwi_dir = PROJECT / "dwi" / s
    dwi = dwi_dir / "data.nii.gz"
    bvals = dwi_dir / "bvals"
    bvecs = dwi_dir / "bvecs"
    mask = dwi_dir / "nodif_brain_mask.nii.gz"
    out = STUDY / s

    if (out / "fit_NDI_modulated.nii.gz").exists() and not FORCE:
        print(f"[{s}] NODDI maps exist")
        continue

    missing = [p.name for p in (dwi, bvals, bvecs, mask) if not p.exists()]
    if missing:
        print(f"[{s}] SKIP: missing {' ', '.join(missing)}")
        continue

    # AMICO empties the output directory when it saves, so the scheme file and
    # the response kernels are kept in a separate working directory.
    work = STUDY / "work" / s
    work.mkdir(parents=True, exist_ok=True)
    out.mkdir(parents=True, exist_ok=True)

    scheme = work / f"{s}.scheme"
    amico.util.fsl2scheme(str(bvals), str(bvecs), str(scheme), bStep=B_STEP)

    ae = amico.Evaluation(str(STUDY), s, str(out))
    ae.set_config("nthreads", int(N_THREADS))
    ae.set_config("BLAS_nthreads", 1)
    ae.set_config("doSaveModulatedMaps", True)
    ae.set_config("doComputeRMSE", True)
    ae.load_data(str(dwi), str(scheme), str(mask), b0_thr=B0_THRESHOLD)

    ae.set_model("NODDI")
    ae.set_config("ATOMS_path", str(work / "kernels")) # after set_model, which resets it
    if D_PAR:
        ae.model.set(float(D_PAR), D_ISO, ae.model.IC_VFs, ae.model.IC_ODs, False)
    ae.generate_kernels(regenerate=True)
    ae.load_kernels()
    ae.fit()
    ae.save_results()
    print(f"[{s}] NODDI maps written (dPar = {ae.model.dPar})")

print(f"DONE -> {STUDY}")

```

08b_build_analysis_csv.py

```

#!/usr/bin/env python3
"""Step 8b. Assemble the analysis files read by the Step 9 scripts.

```

Run in a shell where 00_config.sh has been sourced:

```

source 00_config.sh
python 08b_build_analysis_csv.py

```

Inputs

\$OUT/nodewise/<TRACT>_nodewise_all_subjects.csv	node profiles (Step 8)
\$OUT/nodewise/<TRACT>_tract_stats.csv	cleaned-bundle count and length (Step 6)
\$COVARIATES_CSV	Subject, covariates, outcomes

Output, one file per metric, one row per participant:

```
$OUT/analysis/<TRACT>__<METRIC>__analysis.csv
Subject, covariates, outcomes, Streamline_count, Mean_length_mm, <METRIC>_0 ... <METRIC>_99
"""
import os
import sys
from pathlib import Path

import pandas as pd

def env(name):
    if name not in os.environ:
        sys.exit(f"{name} is not set: run `source 00_config.sh` first")
    return os.environ[name]

OUT = Path(env("OUT"))
TRACT = env("TRACT")

profiles = pd.read_csv(OUT / "nodewise" / f"{TRACT}_nodewise_all_subjects.csv",
                      dtype={"Subject": str})
tract_stats = pd.read_csv(OUT / "nodewise" / f"{TRACT}_tract_stats.csv", dtype={"Subject": str})
covariates = pd.read_csv(env("COVARIATES_CSV"), dtype={"Subject": str})
metrics = [c for c in profiles.columns if c not in ("Subject", "Tract", "Node")]

analysis_dir = OUT / "analysis"
analysis_dir.mkdir(parents=True, exist_ok=True)

for metric in metrics:
    wide = profiles.pivot(index="Subject", columns="Node", values=metric)
    wide.columns = [f"{metric}_{int(node)}" for node in wide.columns]
    bundle = tract_stats[["Subject", "Streamline_count", "Mean_length_mm"]]
    table = covariates.merge(bundle, on="Subject").merge(wide.reset_index(), on="Subject")
    path = analysis_dir / f"{TRACT}__{metric}__analysis.csv"
    table.to_csv(path, index=False)
    print(f"{path.name}: {len(table)} participants, {wide.shape[1]} nodes")

profiled = set(profiles["Subject"])
absent = sorted(profiled - set(covariates["Subject"]))
if absent:
    print(f"profiled but absent from the covariate file: {' '.join(absent)}")
```

09a_tract_models.py

```
#!/usr/bin/env python3
"""Step 9a. Whole-tract and quartile models.
```

Run in a shell where 00_config.sh has been sourced:

```
source 00_config.sh
python 09a_tract_models.py --metric NDI --outcomes memory_accuracy,memory_bias
python 09a_tract_models.py --metric NDI --outcomes memory_accuracy \
    --tracts l_vta_l_hipp,anterior_l_vta_l_hipp
```

The metric is averaged over the whole tract and within quartiles (nodes 0-24, 25-49, 50-74, 75-99). All variables are standardized, so coefficients are standardized betas.

One tract. For each segment: $\text{outcome} \sim \text{metric} + \text{covariates}$ (ordinary least squares).
Quartile p values are corrected across the four quartiles (Benjamini-Hochberg).

Uniformity along the tract: $\text{outcome} \sim \text{mean}(Q1, Q4) + (Q4 - Q1) + \text{covariates}$.
An effect carried by the mean and not by the difference is diffuse, not localized.

Two tracts that share most of their course (--tracts a,b). Both are stacked, two rows per participant, each row with the streamline count and length of its own tract:
 $\text{metric} \sim \text{outcome} + \text{tract} + \text{covariates} + (1 \mid \text{participant})$, fitted by maximum likelihood. The outcome x tract interaction is tested by likelihood ratio (1 df); the main effect is reported from the model without the interaction.

Input: \$OUT/analysis/<TRACT>__<METRIC>__analysis.csv (Step 8b; --analysis-dir overrides)
Output: \$OUT/analysis/<label>__<METRIC>__tract_models.csv

```
"""
import argparse
import os
import sys
import warnings
from pathlib import Path

import pandas as pd
import statsmodels.formula.api as smf
from scipy.stats import chi2
from statsmodels.stats.multitest import multipletests

def env(name):
    if name not in os.environ:
        sys.exit(f"{name} is not set: run `source 00_config.sh` first")
    return os.environ[name]

parser = argparse.ArgumentParser(description="Whole-tract and quartile models")
parser.add_argument("--metric", required=True, help="metric prefix, e.g. NDI")
parser.add_argument("--outcomes", required=True, help="comma-separated outcome columns")
parser.add_argument("--tracts", default=None, help="one tract, or two separated by a comma")
parser.add_argument("--trim", type=int, default=0, help="nodes excluded at each end")
parser.add_argument("--analysis-dir", default=None)
args = parser.parse_args()

METRIC = args.metric
OUTCOMES = args.outcomes.split(",")
TRACTS = (args.tracts or env("TRACT")).split(",")
COVARIATES = env("COVARIATES").split(",")
ANALYSIS = Path(args.analysis_dir) if args.analysis_dir else Path(env("OUT")) / "analysis"
if len(TRACTS) > 2:
    sys.exit("--tracts takes one tract or two")
if not 0 <= args.trim < 25:
    sys.exit("--trim must be between 0 and 24 (nodes excluded at each end)")

SEGMENTS = {"Whole": (args.trim, 100 - args.trim), "Q1": (args.trim, 25), "Q2": (25, 50),
            "Q3": (50, 75), "Q4": (75, 100 - args.trim)}

def q(column):
    """Quote a column name so that any name is valid in a model formula."""
    return f'Q("{column}")'

RHS = " + ".join(q(c) for c in COVARIATES)

def zscore(frame):
    return (frame - frame.mean()) / frame.std(ddof=1)
```

```

def load(tract):
    table = pd.read_csv(ANALYSIS / f"{tract}__METRIC__analysis.csv", dtype={"Subject": str})
    absent = [c for c in COVARIATES + OUTCOMES if c not in table.columns]
    if absent:
        sys.exit(f"{tract}: columns not found: {'', '.join(absent)}")
    return table

def segment_mean(table, segment):
    lo, hi = SEGMENTS[segment]
    return table[[f"{METRIC}_" + node for node in range(lo, hi)]].mean(axis=1)

def single_tract(tract):
    """OLS per segment, FDR across quartiles, and the uniformity contrast."""
    table = load(tract)
    rows = []
    for outcome in OUTCOMES:
        quartile_rows = []
        for segment in SEGMENTS:
            d = table[COVARIATES + [outcome]].assign(metric=segment_mean(table, segment))
            d = d.dropna()
            fit = smf.ols(f"{q(outcome)} ~ metric + {RHS}", zscore(d)).fit()
            row = {"model": "ols", "tract": tract, "segment": segment, "outcome": outcome,
                  "term": "metric", "n": int(fit.nobs), "beta": fit.params["metric"],
                  "statistic": fit.tvalues["metric"], "p": fit.pvalues["metric"]}
            rows.append(row)
            if segment != "Whole":
                quartile_rows.append(row)
        adjusted = multipletests([r["p"] for r in quartile_rows], method="fdr_bh")[1]
        for row, p_fdr in zip(quartile_rows, adjusted):
            row["p_fdr"] = p_fdr

    q1, q4 = segment_mean(table, "Q1"), segment_mean(table, "Q4")
    d = table[COVARIATES + [outcome]].assign(mean_q1q4=(q1 + q4) / 2, diff_q4q1=q4 - q1)
    d = d.dropna()
    fit = smf.ols(f"{q(outcome)} ~ mean_q1q4 + diff_q4q1 + {RHS}", zscore(d)).fit()
    for term in ("mean_q1q4", "diff_q4q1"):
        rows.append({"model": "uniformity", "tract": tract, "segment": "Q1,Q4",
                    "outcome": outcome, "term": term, "n": int(fit.nobs),
                    "beta": fit.params[term], "statistic": fit.tvalues[term],
                    "p": fit.pvalues[term]})

    return rows

def two_tracts(tract_a, tract_b):
    """Mixed model across two tracts; likelihood-ratio test of outcome x tract."""
    tables = {tract: load(tract) for tract in (tract_a, tract_b)}
    rows = []
    for outcome in OUTCOMES:
        for segment in SEGMENTS:
            stacked = pd.concat([
                t[["Subject", outcome] + COVARIATES].assign(metric=segment_mean(t, segment),
                                                            tract=name)
                for name, t in tables.items()], ignore_index=True).dropna()
            numeric = ["metric", outcome] + COVARIATES
            stacked[numeric] = zscore(stacked[numeric])
            with warnings.catch_warnings():
                warnings.simplefilter("ignore")
            main = smf.mixedlm(f"metric ~ {q(outcome)} + tract + {RHS}", stacked,
                              groups=stacked["Subject"]).fit(reml=False)
            full = smf.mixedlm(f"metric ~ {q(outcome)} * tract + {RHS}", stacked,

```

```

                                groups=stacked["Subject"]).fit(reml=False)
    lr = max(2 * (full.llf - main.llf), 0)
    rows.append({"model": "mixed", "tract": f"{tract_a}+{tract_b}", "segment": segment,
                "outcome": outcome, "term": outcome,
                "n": stacked["Subject"].nunique(), "beta": main.params[q(outcome)],
                "statistic": main.tvalues[q(outcome)], "p": main.pvalues[q(outcome)],
                "interaction_p": chi2.sf(lr, 1)})

    return rows

results = []
for tract in TRACTS:
    results += single_tract(tract)
if len(TRACTS) == 2:
    results += two_tracts(*TRACTS)

results = pd.DataFrame(results)
label = "+".join(TRACTS)
path = ANALYSIS / f"{label}__{METRIC}__tract_models.csv"
results.to_csv(path, index=False)
with pd.option_context("display.width", 200, "display.float_format", "{:.3f}".format):
    print(results.to_string(index=False))
print(f"\n-> {path}")

```

09b_nodewise_permutation.R

```

# =====
# Step 9b. Node-wise cluster-extent permutation test (Freedman-Lane)
# =====
# At each node: outcome ~ metric_node + covariates. The residuals of the
# reduced model (covariates only) are permuted, all nodes are refitted, and the
# maximum cluster extent of each permutation forms the null distribution.
# Adapted from a reference implementation used in atlas construction.
#
# Usage (in a shell where 00_config.sh has been sourced):
# Rscript 09b_nodewise_permutation.R <analysis.csv> <outcome> <METRIC>_ <out_dir> <label>
# 1. analysis.csv wide analysis file from Step 8b
# 2. outcome outcome column, e.g. memory_accuracy
# 3. METRIC_ node-column prefix including the underscore, e.g. NDI_
# 4. out_dir directory for the results
# 5. label <tract>__<metric>__<outcome>, used in the output file names
# Environment: COVARIATES (comma-separated columns), N_PERMUTATIONS (default 5000),
# R_PERM_CORES (default: all cores but one)
# Outputs: <label>_nodewise.csv, <label>_clusters.csv, <label>_summary.csv
# =====
suppressPackageStartupMessages({
  library(readr); library(dplyr); library(stringr)
  library(foreach); library(doParallel); library(parallel); library(tibble)
})

args <- commandArgs(trailingOnly = TRUE)
stopifnot(length(args) >= 5)
data_csv <- args[1]
response_col <- args[2]
metric_prefix <- args[3]
out_dir <- args[4]
base <- args[5]

# Covariates: comma-separated column names, exported by 00_config.sh
covariate_env <- Sys.getenv("COVARIATES", unset = "")
if (covariate_env == "") stop("COVARIATES is not set: run `source 00_config.sh` first")

```

```

covariate_cols <- strsplit(covariate_env, ",")[1]

alpha_node      <- 0.05
alpha_familywise <- 0.05
num_permutations <- as.integer(Sys.getenv("N_PERMUTATIONS", unset = "5000"))
rng_seed        <- 123
use_parallel     <- TRUE

dir.create(out_dir, recursive = TRUE, showWarnings = FALSE)

dat <- readr::read_csv(data_csv, show_col_types = FALSE)

# Collect node columns
node_cols <- grep(paste0("^", metric_prefix, "\\d+$"), names(dat), value = TRUE)
if (length(node_cols) == 0) {
  stop("No columns matching ", metric_prefix, "0.. in: ", data_csv)
}
node_idx <- as.integer(sub(paste0("^", metric_prefix), "", node_cols))
ord      <- order(node_idx)
node_cols <- node_cols[ord]; node_idx <- node_idx[ord]
num_nodes <- length(node_cols)

# Numeric casts
dat[[response_col]] <- as.numeric(dat[[response_col]])
for (cc in covariate_cols) dat[[cc]] <- as.numeric(dat[[cc]])

# Drop incomplete
all_model_cols <- c(response_col, covariate_cols, node_cols)
mask <- complete.cases(dat[, all_model_cols])
n_dropped <- sum(!mask)
if (n_dropped) message("Dropping ", n_dropped, " incomplete participants")
dat <- dat[mask, , drop = FALSE]
n_subj <- nrow(dat)
if (n_subj < 5) {
  message("Fewer than 5 complete participants (", n_subj, ") - skipping ", base)
  quit(status = 0)
}

y <- dat[[response_col]]

# Formulas
full_formula <- as.formula(
  paste("y ~ node +", paste(covariate_cols, collapse = " + "))
)
red_formula <- as.formula(
  paste("y ~", paste(covariate_cols, collapse = " + "))
)

# Nodewise fit
fit_node_full <- function(y, node, covariates_df) {
  df0 <- data.frame(y = y, node = node, covariates_df)
  df0 <- df0[complete.cases(df0), ]
  if (nrow(df0) < 3 || sd(df0$node) == 0)
    return(c(Estimate=NA, t=NA, p=NA, df=NA, n=nrow(df0)))
  fit <- lm(full_formula, data = df0)
  sm <- summary(fit)$coefficients
  if (!("node" %in% rownames(sm)))
    return(c(Estimate=NA, t=NA, p=NA, df=fit$df.residual, n=nrow(df0)))
  tval <- unname(sm["node", "t value"])
  est <- unname(sm["node", "Estimate"])
  pval <- 2*pt(-abs(tval), df=fit$df.residual)
  c(Estimate=est, t=tval, p=pval, df=fit$df.residual, n=nrow(df0))
}

```

```

set.seed(rng_seed)
node_stats <- lapply(seq_along(node_cols), function(i) {
  fit_node_full(y, dat[[node_cols[i]]], dat[, covariate_cols, drop=FALSE])
})
node_stats <- do.call(rbind, node_stats)
node_stats_df <- tibble(
  Node=node_idx, Estimate=node_stats[, "Estimate"],
  t_value=node_stats[, "t"], p_value=node_stats[, "p"],
  df=node_stats[, "df"], n=node_stats[, "n"]
)
write_csv(node_stats_df, file.path(out_dir, paste0(base, "_nodewise.csv")))

# Cluster helpers
clusters_from_sig <- function(sig, nodes_numeric) {
  if (!any(sig)) return(list())
  idx <- which(sig); cls <- list(); run <- c(nodes_numeric[idx[1]])
  if (length(idx) > 1) for (k in 2:length(idx)) {
    if (nodes_numeric[idx[k]] == nodes_numeric[idx[k-1]] + 1)
      run <- c(run, nodes_numeric[idx[k]])
    else { cls[[length(cls)+1]] <- run; run <- c(nodes_numeric[idx[k]]) }
  }
  cls[[length(cls)+1]] <- run; cls
}
max_cluster_size_from_sig <- function(sig, nodes_numeric) {
  cls <- clusters_from_sig(sig, nodes_numeric)
  if (length(cls)==0) 0L else max(vapply(cls, length, 1L))
}

sig_mask <- !is.na(node_stats_df$p_value) & node_stats_df$p_value < alpha_node
obs_clusters <- clusters_from_sig(sig_mask, node_stats_df$Node)
obs_sizes <- vapply(obs_clusters, length, 1L)
obs_max_size <- if (length(obs_sizes)) max(obs_sizes) else 0L
num_sig_nodes <- sum(sig_mask)
num_clusters <- length(obs_clusters)

# Freedman-Lane
df_red_global <- data.frame(y=y, dat[, covariate_cols, drop=FALSE])
fit_red <- lm(red_formula, data=df_red_global)
yhat_red <- fitted(fit_red); resid_red <- resid(fit_red)

analysable <- vapply(seq_along(node_cols), function(i)
  sd(dat[[node_cols[i]]]) > 0, logical(1))

# All permutations are drawn once, here, so that the null distribution is identical
# for any number of cores and any scheduling of the parallel workers.
set.seed(rng_seed)
perm_mat <- vapply(seq_len(num_permutations), function(k) sample.int(n_subj),
  integer(n_subj))

perm_fun <- function(.perm) {
  perm_idx <- perm_mat[, .perm]
  y_perm <- yhat_red + resid_red[perm_idx]
  p_perm <- rep(1, num_nodes)
  for (i in which(analysable)) {
    df_perm <- data.frame(y=y_perm, node=dat[[node_cols[i]]],
      dat[, covariate_cols, drop=FALSE])
    fit <- lm(full_formula, data=df_perm)
    sm <- summary(fit)$coefficients
    if ("node" %in% rownames(sm)) {
      tval <- sm["node", "t value"]
      p_perm[i] <- 2*pt(-abs(tval), df=fit$df.residual)
    }
  }
}

```

```

    }
    max_cluster_size_from_sig(p_perm < alpha_node, node_idx)
  }

env_cores <- suppressWarnings(as.integer(Sys.getenv("R_PERM_CORES", unset = NA)))
if (!is.na(env_cores) && env_cores > 0L) {
  cores <- env_cores
} else {
  cores <- max(1L, parallel::detectCores() - 1L)
}

if (cores > 1L) {
  cl <- parallel::makeCluster(cores); registerDoParallel(cl)
  perm_max_sizes <- foreach(perm = 1:num_permutations, .combine = c,
    .packages = "stats") %dopar% perm_fun(perm)
  parallel::stopCluster(cl)
} else {
  perm_max_sizes <- vapply(1:num_permutations, perm_fun, integer(1))
}

# Cluster p value: proportion of permutations whose largest cluster is at least as long.
# The extent threshold is the smallest extent with p <= alpha. Extents are whole
# numbers, so this is one node above the (1 - alpha) quantile; a cluster exactly at the
# quantile has p > alpha and does not pass.
cluster_p_from_size <- function(size) mean(perm_max_sizes >= size)
extent_threshold <- as.integer(quantile(perm_max_sizes, probs = 1 - alpha_familywise,
  type = 1)) + 1L

if (num_clusters > 0) {
  all_clusters_df <- do.call(rbind, lapply(seq_along(obs_clusters), function(k) {
    nodes <- obs_clusters[[k]]; idx_in_table <- match(nodes, node_stats_df$Node)
    t_vals <- node_stats_df$t_value[idx_in_table]
    ests <- node_stats_df$Estimate[idx_in_table]
    tibble(
      ClusterID=k, Size=length(nodes), StartNode=min(nodes), EndNode=max(nodes),
      Nodes=paste(nodes, collapse=","),
      MeanTValue=mean(t_vals, na.rm=TRUE),
      Direction=ifelse(mean(t_vals, na.rm=TRUE)>0,"Positive","Negative"),
      MaxAbsTValue=t_vals[which.max(abs(t_vals))],
      MaxAbsTNode=nodes[which.max(abs(t_vals))],
      MeanEstimate=mean(ests, na.rm=TRUE),
      ClusterPValue=cluster_p_from_size(length(nodes)),
      ExtentThresholdNodes=extent_threshold,
      PassExtentThreshold=cluster_p_from_size(length(nodes)) <= alpha_familywise
    )
  }))
} else {
  all_clusters_df <- tibble(
    ClusterID=integer(), Size=integer(), StartNode=integer(), EndNode=integer(),
    Nodes=character(), MeanTValue=double(), Direction=character(),
    MaxAbsTValue=double(), MaxAbsTNode=integer(),
    MeanEstimate=double(),
    ClusterPValue=double(), ExtentThresholdNodes=integer(), PassExtentThreshold=logical()
  )
}
write_csv(all_clusters_df, file.path(out_dir, paste0(base, "_clusters.csv")))

summary_df <- tibble(
  Outcome=response_col,
  MetricPrefix=metric_prefix,
  Covariates=paste(covariate_cols, collapse=","),
  N_subjects=n_subj, N_dropped=n_dropped,
  NodesTested=num_nodes,

```

```

    NodewiseAlpha=alpha_node, FamilywiseAlpha=alpha_familywise,
    NumPermutations=num_permutations,
    NumNodewiseSignificant=num_sig_nodes,
    NumClustersFormed=num_clusters,
    ObservedMaxClusterSize=obs_max_size,
    ExtentThresholdNodes=extent_threshold,
    NumClustersPassingExtent=sum(all_clusters_df$PassExtentThreshold, na.rm=TRUE)
)
write_csv(summary_df, file.path(out_dir, paste0(base, "_summary.csv")))
message("Done: ", base, " - clusters=", num_clusters,
        " sig nodes=", num_sig_nodes, " ext_thr=", extent_threshold)

```

09c_stack_for_explorer.py

```

#!/usr/bin/env python3
"""Step 9c. Stack the node-wise results into the file read by the Explorer.

    python 09c_stack_for_explorer.py /path/to/permutation/results

Expects the three outputs of 09b_nodewise_permutation.R for every analysis, named
<tract>__<metric>__<outcome>_nodewise.csv, _clusters.csv and _summary.csv.

Output: results_long.csv in the same directory, one row per node, with columns
    outcome, tract, metric, node, t, p, hemisphere, N, covariates,
    extent_threshold, cluster_p, passed
hemisphere is L or R when the tract name contains the token l or r (l_vta_l_hipp,
anterior_r_vta_r_hipp) and is otherwise empty; cluster_p and passed
describe the largest cluster of the analysis that met the extent threshold.
"""
import sys
from pathlib import Path

import pandas as pd

results_dir = Path(sys.argv[1] if len(sys.argv) > 1 else ".")
rows = []

for summary_file in sorted(results_dir.glob("*_summary.csv")):
    label = summary_file.name[:-len("_summary.csv")]
    parts = label.split("__", 2)
    if len(parts) != 3:
        print(f"skipped {summary_file.name}: label is not <tract>__<metric>__<outcome>")
        continue
    tract, metric, outcome = parts

    summary = pd.read_csv(summary_file).iloc[0]
    nodewise = pd.read_csv(results_dir / f"{label}_nodewise.csv")
    clusters = pd.read_csv(results_dir / f"{label}_clusters.csv")
    passing = clusters[clusters["PassExtentThreshold"].astype(str).str.upper() == "TRUE"]

    tokens = tract.lower().split("_")
    hemisphere = next((t.upper() for t in tokens if t in ("l", "r")), "")

    for node in nodewise.itertuples():
        rows.append({
            "outcome": outcome, "tract": tract, "metric": metric,
            "node": int(node.Node), "t": round(node.t_value, 4), "p": round(node.p_value, 5),
            "hemisphere": hemisphere, "N": int(summary["N_subjects"]),
            "covariates": summary["Covariates"],
            "extent_threshold": int(summary["ExtentThresholdNodes"]),
            "cluster_p": passing["ClusterPValue"].min() if len(passing) else "",

```

```

        "passed": int(len(passing) > 0),
    })

if not rows:
    sys.exit(f"no *_summary.csv files found in {results_dir}")
out = results_dir / "results_long.csv"
pd.DataFrame(rows).to_csv(out, index=False)
print(f"wrote {out} ({len(rows)} rows, {len(rows) // 100} analyses)")

```

00b_fod_estimation.sh

```

#!/bin/bash
# =====
# Step 0b. Fibre orientation distributions (multi-shell multi-tissue CSD)
# =====
# Run once, before Step 5, when preprocessing ended at the tensor.
#   Phase 1 (per participant) response functions: dwi2response dhollander
#   Phase 2 (group)         average response functions: responsemean
#   Phase 3 (per participant) dwi2fod msmt_csd, then mtnormalise
# Single-shell data support two tissues only: remove the GM response and GM FOD
# arguments from the dwi2fod and mtnormalise calls.
# =====
source "${dirname "$0"}/00_config.sh"
start_log "$0"
read_subjects

estimate_response() {
    local s=$1
    local d="$PROJECT/dwi/$s"
    local f
    for f in data.nii.gz bvals bvecs nodif_brain_mask.nii.gz; do
        if [ ! -f "$d/$f" ]; then
            echo "!! $s missing $d/$f"
            return
        fi
    done
    if [ "$FORCE" != 1 ] && [ -f "$d/wm_response.txt" ]; then
        echo "== $s response functions exist"
        return
    fi
    mrconvert "$d/data.nii.gz" "$d/dwi.mif" -fslgrad "$d/bvecs" "$d/bvals" -force -quiet
    mrconvert "$d/nodif_brain_mask.nii.gz" "$d/mask.mif" -force -quiet
    dwi2response dhollander "$d/dwi.mif" \
        "$d/wm_response.txt" "$d/gm_response.txt" "$d/csf_response.txt" \
        -mask "$d/mask.mif" -force -quiet
    echo ">> $s response functions"
}

estimate_fod() {
    local s=$1
    local d="$PROJECT/dwi/$s"
    local g="$PROJECT/dwi"
    if [ "$FORCE" != 1 ] && [ -f "$d/wm_fod_norm.mif" ]; then
        echo "== $s FOD exists"
        return
    fi
    dwi2fod msmt_csd "$d/dwi.mif" \
        "$g/group_wm_response.txt" "$d/wm_fod.mif" \
        "$g/group_gm_response.txt" "$d/gm_fod.mif" \
        "$g/group_csf_response.txt" "$d/csf_fod.mif" \
        -mask "$d/mask.mif" -nthreads "$NTHREADS" -force -quiet
}

```

```

# The normalized WM FOD marks a finished participant, so it is written under a
# temporary name and renamed only when mtnormalise succeeds.
if mtnormalise \
    "$d/wm_fod.mif" "$d/wm_fod_norm.partial.mif" \
    "$d/gm_fod.mif" "$d/gm_fod_norm.mif" \
    "$d/csf_fod.mif" "$d/csf_fod_norm.mif" \
    -mask "$d/mask.mif" -force -quiet; then
    mv -f "$d/wm_fod_norm.partial.mif" "$d/wm_fod_norm.mif"
    echo ">> $s FOD"
else
    rm -f "$d/wm_fod_norm.partial.mif"
    echo "!! $s FOD estimation failed or was interrupted"
fi
}

# Phase 1
for s in "${SUBJECTS[@]}; do
    estimate_response "$s" &
    throttle "$MAXJOBS"
done
wait_for_jobs

# Phase 2
for tissue in wm gm csf; do
    responsemean "$PROJECT"/dwi/*/$(tissue)_response.txt \
        "$PROJECT"/dwi/group_$(tissue)_response.txt" -force -quiet
done
echo "== group response functions written"

# Phase 3 (memory-intensive: $FOD_JOBS participants at a time)
for s in "${SUBJECTS[@]}; do
    estimate_fod "$s" &
    throttle "$FOD_JOBS"
done
wait_for_jobs

# Audit
printf "\nSubject\twm_fod_norm\n"
for s in "${SUBJECTS[@]}; do
    printf "%s\t%s\n" "$s" "$(present "$PROJECT"/dwi/$s/wm_fod_norm.mif)"
done

```